

On Latent Predictive Residuals and Automated Threshold Calibration for Multivariate Time-Series Anomaly Detection: An Empirical Investigation

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Abstract—Unsupervised anomaly detection in multivariate telemetry requires setting decision thresholds automatically via statistical calibration methods, such as Extreme Value Theory (EVT / SPOT), without access to ground-truth labels. While deep reconstructive architectures such as TimesNet and TranAD achieve competitive continuous ranking performance (PR-AUC 0.29–0.34), their observation-space residuals exhibit severe tail heaviness under sensor jitter. When calibrated under automated streaming thresholds, these heavy tails trigger excessive false alarms, with nominal false-positive rates reaching 24–30%. We investigate whether forecasting in abstract representation space via Joint-Embedding Predictive Architectures (JEPAs) produces prediction residuals that are better conditioned for automated thresholding. Alongside an unconstrained latent predictor (TS-JEPA), we evaluate three candidate physics-inspired regularizers: Koopman spectral entropy, Reynolds second-moment fluctuation covariance, and curl-free potential flow gradients. We benchmark these models across 33 matched continuous telemetry streams from seven physical datasets using a strict held-out calibration protocol (chronological 65/35 train/hold-out split with a 64-step isolation gap, multiple seeds, and matched-capacity controls). Our empirical evaluation yields four main findings. First, latent prediction lowers the measured nominal false-positive rate under SPOT from 24–30% to 5.2–7.9% (paired difference -0.154 against TimesNet, $p = 0.008$; -0.157 against TranAD, $p = 0.001$). Second, overall Point-F1 (0.254 for TS-JEPA vs. 0.207 for TimesNet, $p = 0.55$) and PR-AUC (0.333 vs. 0.336, $p = 0.30$) are not statistically distinguishable from baselines, and under strictly causal streaming (zero lookahead), JEPA event recall falls from 32.9% to 10.7%, indicating a conservative operating trade-off rather than universal dominance. Third, controlled ablations on fixed architectures show that none of the three physical regularizers improves Point-F1 beyond seed noise ($+0.001$ for Operator-Entropy, 0.000 for Reynolds-Stress, and $+0.006$ for Potential-Flow; $p \geq 0.58$), indicating that measured performance differences arise from shared neural modules rather than physical inductive priors. Fourth, automated EVT calibration systematically fails to reach its nominal design risk ($q = 10^{-3}$) on non-stationary telemetry, performing similarly to empirical percentiles, and kurtosis-adaptive heuristics fail to correct this drift. We release our complete multi-seed benchmark code and evaluation logs to support reproducible research in streaming anomaly detection.

Index Terms—Multivariate Time-Series, Anomaly Detection, Joint-Embedding Predictive Architecture (JEPA), Residual Conditioning, Threshold Calibration, Extreme Value Theory, Empirical Benchmark.

I. INTRODUCTION

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments (such as the NASA Soil Moisture Active Passive (SMAP) satellite, Mars Science Laboratory (MSL) rovers, industrial IoT facilities, and wearable biometric sensor arrays), dozens to hundreds of continuous telemetry channels concurrently track electrical power buses, thermal gradients, mechanical vibrations, and kinematics [3]. When sensor faults or operational breakdowns emerge at the edge, monitoring systems must identify dynamical anomalies before hardware damage occurs [4].

A. Challenges in Modeling Multi-Sensor Telemetry

From a physical perspective, multi-sensor telemetry reflects the state trajectory of an underlying continuous physical system $\dot{x}(t) = f(x(t))$. However, attempting to model dynamical equations directly in raw observation space faces four core challenges on edge devices. First, estimating instantaneous time derivatives $\dot{x} \approx \frac{\Delta x}{\Delta t}$ directly on discrete, noise-corrupted sensor streams amplifies high-frequency measurement noise, destabilizing numerical trajectory tracking. Second, remote sensing telemetry captures only a noisy surface projection of unmeasured thermodynamic, electrical, and environmental states, violating full-state observability. Third, in large sensor arrays ($K \geq 25$, such as NASA SMAP with $K = 25$ and MSL with $K = 55$), cross-channel dependencies and non-linear interactions cannot be adequately represented by rigid functional forms in raw signal space. Finally, edge IoT and satellite systems undergo discrete operational mode transitions (e.g., orbital day/night thermal cycling, thruster firings, Parkinsonian gait freezing, cloud server garbage collection) that cause continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Deep learning architectures provide data-driven alternatives to explicit equation modeling, yet they face three primary limitations. First, generative models (e.g., autoencoders, VAEs,

masked Transformers) minimize point-level L_1/L_2 errors in raw signal space [3], [5], [6]. Because reconstruction loss penalizes every discrepancy equally, autoencoders allocate capacity to fitting high-frequency sensor jitter rather than learning nominal dynamical transitions. Second, contrastive frameworks such as Neural Contextual Anomaly Detection (NCAD) [7] train encoders via Contextual Outlier Exposure (COE) using heuristic synthetic perturbations (e.g., Gaussian noise spikes, sequence splicing). However, real physical faults manifest as subtle deviations from nominal dynamics; synthetic injections distort embedding geometry, inducing artificial decision boundaries that generate false alarms on continuous telemetry. Third, in high-dimensional observation space, even after per-channel standardization, Euclidean reconstruction errors ($\|x - \hat{x}\|_2^2 = \sum_k (x_k - \hat{x}_k)^2$) accumulate stochastic residual variance across dozens of irrelevant or noisy sensors. Under automated streaming thresholding such as Peaks-Over-Threshold (SPOT), this residual accumulation yields heavy, unstable tail distributions that distort extreme-value calibrators, triggering false alarm flooding in operational telemetry.

C. Investigating Latent Prediction and Inductive Priors

To investigate whether operating in latent representation space mitigates observation-space noise drowning without relying on synthetic contrastive perturbations, we study Joint-Embedding Predictive Architectures (JEPAs) for multivariate telemetry. Rather than reconstructing raw, jitter-corrupted sensor observations $\hat{x} \approx x$, a JEPA maps past observations x_{ctx} into an abstract representation space and predicts target representations z_{tgt} directly [8], [9]. By predicting in latent space, the architecture discards high-frequency stochastic sensor noise that lacks temporal predictability, potentially conditioning prediction residuals for automated tail calibration.

Alongside an unconstrained latent predictor (TS-JEPA), we formalize and test three candidate physics-inspired regularizers hypothesized to stabilize representation dynamics. The first formulation, *Operator-Entropy JEPA*, regularizes the learned latent Koopman transition matrix via spectral entropy to promote high-rank modal expressivity. The second, *Reynolds-Stress JEPA*, adapts hydrodynamic Reynolds decomposition ($z = \bar{z} + z'$) to penalize discrepancies in cross-coordinate second-moment fluctuation covariance. The third, *Potential-Flow JEPA*, parameterizes latent transitions as gradients of a scalar potential ($\hat{v} = -\nabla\Phi$), enforcing curl-free dynamics. To isolate channel activity and decouple multi-scale frequencies, these architectures incorporate Causal Coordinate Saliency Gating and Littlewood-Paley Dyadic Wavelet Shells, calibrated via Peaks-Over-Threshold (SPOT).

D. Contributions

This work makes four principal contributions:

1) Quantifying the Latent Residual Conditioning Effect: We demonstrate that forecasting telemetry states in abstract representation space rather than reconstructing raw observation waveforms produces residual distributions with lower tail heaviness, lowering measured nominal false-positive rates under SPOT from 24–30% to 5.2–7.9% (paired difference

−0.154 against TimesNet, $p = 0.008$; −0.157 against TranAD, $p = 0.001$). However, Point-F1 and PR-AUC remain statistically indistinguishable from reconstructive baselines, and strictly causal streaming incurs an event recall penalty (10.7% vs. 32.9%), showing that latent prediction provides a conservative, high-precision operating point rather than universal dominance.

2) Rigorous Evaluation of Physics-Inspired Inductive Priors: We evaluate the three physics-inspired architectures (Operator-Entropy JEPA, Reynolds-Stress JEPA, and Potential-Flow JEPA) against unconstrained TS-JEPA. Across matched-capacity controls on fixed neural architectures, switching off the physical regularizers yields no statistically significant difference in Point-F1 (+0.001 for entropy, 0.000 for stress, and +0.006 for potential flow; $p \geq 0.58$), documenting rigorous null results for these physical regularizers on real telemetry.

3) Empirical Analysis of Threshold Calibration Failures: We document that automated EVT/SPOT calibration systematically fails to achieve its theoretical design risk ($q = 10^{-3}$) on unlabelled non-stationary telemetry, resulting in empirical false-positive rates of 5–18% due to distribution drift. We further demonstrate that kurtosis-adaptive heuristics fail to correct this drift, performing comparably to empirical quantile baselines.

4) Open-Source, Reproducible Multi-Seed Benchmark: We establish a fully reproducible benchmark across 33 matched telemetry streams from seven physical datasets evaluated under a strict held-out calibration protocol (chronological 65/35 train/held-out split with a 64-step gap, up to three seeds, paired bootstrap confidence intervals, and Wilcoxon signed-rank tests). All code, checkpoints, and evaluation logs are released open-source.

The remainder of this paper is structured as follows. Section II reviews related work in reconstructive, contrastive, and joint-embedding anomaly detection. Section III details the multi-sensor telemetry benchmark suite and the held-out partition protocol. Section IV formalizes the JEPA formulations, causal saliency gating, dyadic wavelets, and EVT calibration. Section V establishes the experimental setup and evaluation protocol. Section VI evaluates architectural components and inductive regularizer controls. Section VII presents quantitative benchmark results, threshold calibration analyses, and computational profiles. Finally, Section VIII concludes the paper.

II. RELATED WORK

Unsupervised time-series anomaly detection spans dynamical equation discovery, reconstructive generative modeling, contrastive representation learning, and predictive world models. This section reviews these approaches and positions our Physics-Inspired JEPA framework.

A. Dynamical Systems Modeling and Koopman Operators

The physical modeling of multi-sensor telemetry has long sought to capture system behavior as continuous dynamical trajectories governed by differential equations $\dot{x}(t) = f(x(t))$ [10]. One prominent approach, Neural Ordinary Differential

Equations (Neural ODEs) [11], parameterizes continuous velocity fields using neural networks $\frac{dx}{dt} = f_\theta(x(t), t)$ and integrates trajectories numerically. While Neural ODEs learn flexible vector fields, fitting them directly in raw observation space forces numerical solvers to integrate through high-frequency stochastic sensor noise, leading to numerical stiffness, slow training, and sensitivity to measurement jitter.

In parallel, Koopman operator theory and Dynamic Mode Decomposition (DMD) [12]–[14] analyze non-linear dynamical systems by learning linear transition operators that act on observable functions. Modern deep Koopman frameworks (e.g., Koopa [14]) learn observable embeddings for non-stationary forecasting. In anomaly detection, physical disruptions manifest as spectral decoherence or sudden changes in modal dissipation. Rather than operating on uncalibrated raw signals, our framework parameterizes latent linear transitions regularized by Koopman spectral entropy during training directly in representation space.

B. Reconstruction-Based Detection

The predominant deep learning approach in time-series anomaly detection relies on learning nominal patterns through raw signal reconstruction. Hundman et al. [3] introduced LSTM-based prediction coupled with non-parametric dynamic thresholding for NASA spacecraft telemetry (SMAP/MSL). Modern architectures incorporate specialized attention and frequency-domain transformations. For example, Anomaly Transformer [5] introduces Anomaly-Attention to quantify the association discrepancy between local prior Gaussian kernel associations and global series self-attention. TimesNet [6] decomposes 1D multivariate time series into 2D temporal variations using Fast Fourier Transform (FFT) periodicity discovery and 2D Inception blocks. Similarly, TranAD [15] employs a two-phase adversarial Transformer with sub-series attention and focus scoring to amplify subtle reconstruction errors.

Despite their empirical popularity, reconstructive methods share a fundamental vulnerability: signal-level noise memorization. Because loss functions optimize point-level L_1 or L_2 differences in raw signal space, reconstructive networks are biased toward fitting high-frequency stochastic noise rather than learning invariant dynamical laws. In addition, during sustained anomalies, corrupted observations contaminate the context window, causing the reconstructor to adapt and reconstruct the anomaly as the nominal state.

C. Contrastive Learning and Representation-Based Detection

To avoid reconstructing raw noise, contrastive methods learn discriminative representations by separating nominal sequences from anomalous perturbations. Neural Contextual Anomaly Detection (NCAD) [7] uses Temporal Convolutional Networks (TCNs) trained with Contextual Outlier Exposure (COE), injecting artificial synthetic anomalies (Gaussian noise, point spikes, trend shifts) into training windows to create negative pairs. However, synthetic injections distort the physical dynamical manifold and fail to generalize to subtle real-world

physical faults (such as gradual thermal dissipation or Parkinsonian gait freezing). Alternatively, DCdetector [16] replaces reconstruction with multi-scale dual-attention contrastive representation learning, maximizing cross-scale representation consistency. While DCdetector eliminates artificial injection heuristics, contrastive objectives require careful scale selection and representation pooling.

D. Self-Supervised Time-Series Representation Learning and JEPAs

Joint-Embedding Predictive Architectures (JEPA), originally proposed by LeCun [8] and demonstrated in computer vision by I-JEPA [9] and V-JEPA [17], predict target representations from context in latent space. Non-contrastive regularizers such as Barlow Twins [18], W-MSE [19], and VICReg [20] prevent representation collapse without negative mining.

Recent works have extended self-supervised and predictive representation learning principles to time-series sequences. TS2Vec [21] learns multi-scale representations through contextual and temporal contrasting, while Eldele et al. [22] enforce temporal-contextual cross-view consistency. Systematic surveys by Zhang et al. [23] document the emergence of self-supervised representation architectures across multi-sensor streams. However, existing time-series representation frameworks lack explicit physical dynamical priors, leaving latent trajectories unanchored to physical conservation and spectral dissipation principles.

E. Physics-Inspired Deep Learning and Structured Inductive Biases

Physics-Informed Neural Networks (PINNs) [24], alongside Hamiltonian [25] and Lagrangian Neural Networks [26], integrate dynamical principles into neural architectures. However, while Hamiltonian networks enforce strict energy conservation for closed mechanical systems, standard PINNs face two primary limitations in high-dimensional sensor telemetry. First, jointly optimizing data fidelity and PDE residual loss leads to stiff, competing gradient dynamics, often causing training instability or failure to converge in high noise regimes [27]. Second, complex IoT and aerospace telemetry (e.g., SMAP, MSL, and cyber-physical water networks) rarely follow a single closed-form PDE whose analytical derivatives can be evaluated across dozens of uncalibrated sensors. Rather than enforcing analytical differential equations through auxiliary training penalties, our framework incorporates physical conservation, turbulent transport, and spectral dissipation principles directly into latent representations as inductive regularizers, avoiding multi-loss balancing conflicts.

F. Positioning of the Proposed Framework

The proposed Physics-Inspired JEPA framework bridges joint-embedding predictive learning and physical domain mechanics through four distinct structural designs. First, by forecasting future representations z_{tgt} rather than raw observations $\hat{x}$, our models circumvent high-frequency noise drowning and sensor jitter memorization that hobble reconstructive baselines

(TimesNet, TranAD). Second, we investigate three physics-inspired inductive formulations: Operator-Entropy (Koopman spectral dissipation), Reynolds-Stress (turbulent momentum transport), and Potential-Flow (conservative irrotational field theory), addressing cyclic, turbulent, and conservative physical dynamics. Third, we incorporate Causal Coordinate Saliency Gating with zero target leakage to prevent active channel dilution in high-dimensional arrays, and Littlewood-Paley Dyadic Wavelets to isolate transient impulses from low-frequency operational drift. Finally, we deploy Extreme Value Theory (EVT / SPOT) [28] modulated by dynamic sample kurtosis, establishing decision boundaries on operational telemetry without transductive test-label tuning.

III. MULTI-SENSOR IoT AND REMOTE SENSING BENCHMARK SUITE

To evaluate our Physics-Inspired JEPA framework across diverse edge, cyber-physical, and orbital sensing dynamics, our benchmark suite comprises 11 datasets spanning 45 strictly matched continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [29], [30], TimeEval [31], and mTSBench [32]. We organize these streams into four physical sensing domains:

A. Domain 1: Satellite and Spacecraft Telemetry (13 Streams)

This domain covers high-reliability orbital and deep-space telemetry. The *NASA SMAP Earth Observation Satellite* [3] (6 evaluated streams: A-1, A-2, A-7, D-3, P-3, E-1) provides continuous telemetry values alongside one-hot command vectors monitoring power, thermal, and attitude control subsystems. The *NASA MSL Curiosity Rover* [3] (6 evaluated streams: C-1, C-2, D-14, D-15, D-16, M-1) tracks robotic actuator, joint drive, and environmental streams during planetary surface operations. Finally, the *SWAN-SF Space Weather Telemetry* [33] (1 stream) provides solar photospheric vector magnetogram telemetry from NASA SDO/HMI, tracking active region magnetic field perturbations during solar eruptive events.

B. Domain 2: Wearable Biometrics and Kinematics (6 Streams)

This domain evaluates human kinematic telemetry via the *Daphnet Wearable Body Sensor Network* [29] (6 evaluated streams: S01R01E1 to S03R02E0). Telemetry consists of 64 Hz tri-axial accelerometry recorded from wearable sensor nodes placed on the shank, thigh, and trunk to track Freezing-of-Gait (FoG) motor breakdown events in Parkinson's disease patients. At 64 Hz, the suspect horizon $S = 64$ corresponds to a 1.0-second lookahead window.

C. Domain 3: Industrial Processes and Cyber-Physical Systems (16 Streams)

This domain covers continuous industrial processes and fluid transportation. The *GHL Gasoil Heating Loop* (14 benchmark streams) captures thermal and fluid process loop telemetry across temperature, flow rate, and valve pressure dynamics.

The *Genesis Automation Testbed* (1 stream) tracks manufacturing automation telemetry from an industrial pick-and-place demonstrator during mechanical handling faults. The *GECCO Water Distribution IoT* (1 stream) monitors physical water distribution infrastructure through continuous chemical and hydrological measurements at 1-minute resolution (where $S = 64$ represents a 64-minute operational horizon).

D. Domain 4: Smart Infrastructure and Environmental Sensing (10 Streams)

This domain evaluates urban infrastructure and cyber-physical networks. The *Room-Occupancy Environmental IoT* dataset [34] (2 streams) records ambient environmental conditions (CO₂, light, temperature, humidity) tracking room-occupancy state switches. The *CICIDS Edge Network Telemetry* (6 streams) captures network flow dynamics, tracking gateway protocol shifts and security anomalies including DDoS bursts and port scans. The *Callt2 Building People-Flow Telemetry* [30] (1 stream) records continuous pedestrian counts via optical beam sensors at the UC Irvine Calit2 facility entrance. Finally, the *Metro Highway Traffic* dataset (1 stream) tracks interstate highway traffic volume on Minnesota Interstate I-94, where anomalies correspond to major traffic disruption and incident events.

Each benchmark stream provides nominal training sequences and testing sequences containing labeled ground-truth anomalies. Continuous telemetry channels are standardized via training statistics ($\mu_{\text{train}}, \sigma_{\text{train}}$), flooring standard deviations at $\max(\sigma, 10^{-2})$ to prevent division by zero during flatline operational phases. While mission archives contain larger entity counts (the complete SMAP archive includes 55 telemetry entities and MSL includes 27 entities across subsystems), our benchmark evaluates standardized subsets studied in the telemetry anomaly detection literature [3], [31] to ensure strict matching and reproducible runtime conditions across all baselines.

To eliminate circular in-sample calibration and measure out-of-sample false-alarm rates under automated thresholding, we enforce a strict held-out calibration protocol. For each stream, the nominal training sequence is partitioned chronologically: the first 65% is used to train model representations; the remaining 35% is reserved for held-out evaluation, split into a calibration half (used to fit extreme-value thresholds without seeing test anomalies) and an evaluation half (separated by a 64-step isolation gap) on which nominal false-positive rates (FPR) are measured on data the threshold never saw. This multi-seed held-out protocol completes across 33 streams spanning seven distinct physical datasets: 6 Daphnet (wearable kinematics), 14 GHL (fluid transport), 4 MSL (planetary rovers), 6 SMAP (Earth observation satellites), Genesis (manufacturing automation), Metro (highway traffic), and SWAN (solar space weather). Streams with nominal blocks too short to support a calibration and an evaluation half with an isolation gap (Callt2, Room-Occupancy, and two MSL streams) or where multi-seed stride-1 bootstrap evaluation over 10^6 steps was computationally prohibitive (GECCO, CICIDS) are documented in Table I.

Table I provides the technical specifications across the benchmark suite, totaling 2,079,507 nominal training timesteps and 2,986,574 test timesteps (over 5.06 million observations). We explicitly document the physical domain, channel dimension K , sampling resolution, sequence lengths, anomaly contamination ratio $\pi\%$, anomaly failure modes, and partition protocols to ensure experimental reproducibility.

IV. METHODOLOGY: THE PHYSICS-INSPIRED GEOMETRIC JEPA TRIAD

This section formalizes the mathematical formulation and inductive biases of the proposed framework.

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T \times K}$ denote a continuous multivariate time-series sequence consisting of K observed sensor variables over T timesteps. The sequence is partitioned into historical context $x_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ and suspect/target horizon $x_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$, where C and S denote the respective temporal horizons.

Existing deep learning baselines (e.g., TimesNet, TranAD, Anomaly Transformer) frame anomaly detection as pointwise reconstruction:

$$\min_{\theta} \mathbb{E} [\|\mathbf{X} - \mathcal{D}_{\theta}(\mathcal{E}_{\theta}(\mathbf{X}))\|_F^2]. \quad (1)$$

In complex multi-sensor telemetry, ambient high-frequency stochastic jitter dominates the Frobenius norm, causing models to waste representational bandwidth reconstructing sensor noise rather than modeling system state dynamics (leading to noise drowning).

To overcome this, we adopt the Joint-Embedding Predictive Architecture (JEPA) framework [8]. A context encoder E_{θ} maps $x_{\text{ctx}} \rightarrow z_{\text{ctx}} \in \mathbb{R}^D$, while an Exponential Moving Average (EMA) momentum target encoder E_{ϕ} maps $x_{\text{tgt}} \rightarrow \hat{z}_{\text{tgt}} \in \mathbb{R}^D$, updated via:

$$\phi_k \leftarrow m_k \phi_{k-1} + (1 - m_k) \theta_k, \quad m_k \in [0.996, 0.9995]. \quad (2)$$

The core objective is to predict target representations in latent space under physical dynamical constraints.

B. The Physics-Inspired JEPA Triad

Rather than relying on unconstrained Euclidean regression, we incorporate physical domain regularizers through three complementary geometric formulations:

1) *Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion)*: In dynamical systems theory, non-linear trajectories can be analyzed via linear operators acting on observable functions (Koopman operator theory [?], [?]). In our architecture, the encoder maps the observation window to a finite-dimensional latent observable vector $z \in \mathbb{R}^D$. We parameterize the predictive transition between context and target latent states via a learned transition operator $\mathbf{K} \in \mathbb{R}^{D \times D}$:

$$\hat{z}_{\text{tgt}} = \mathbf{K} z_{\text{ctx}}. \quad (3)$$

To prevent the operator from collapsing onto a degenerate low-rank subspace (dimensional collapse) or degenerating to

zero (energy collapse), we formulate an operator-entropy regularizer that decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \dots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, with corresponding eigenvalues $\lambda_i = \sigma_i^2$ of the symmetric Gram matrix $\mathbf{K}\mathbf{K}^T$. The total operator Frobenius energy is $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_{i=1}^D \lambda_i$. For any non-zero operator ($\mathcal{E}(\mathbf{K}) > 0$), we define the scale-free spectral density distribution $\mathbf{p} = [p_1, \dots, p_D]^T$ over latent modes:

$$p_i = \frac{\lambda_i(\mathbf{K}\mathbf{K}^T)}{\sum_{j=1}^D \lambda_j(\mathbf{K}\mathbf{K}^T)} = \frac{\sigma_i^2}{\|\mathbf{K}\|_F^2}, \quad \sum_{i=1}^D p_i = 1, \quad p_i \geq 0. \quad (4)$$

We omit isotropic diagonal loading ($\frac{\epsilon}{D}\mathbf{I}$) from the Gram matrix before normalization; adding it would cause $\mathbf{K} \rightarrow \mathbf{0}$ to yield an artificial identity spectrum with maximal entropy, spuriously rewarding total collapse. Instead, we evaluate the Von Neumann spectral entropy directly on the scale-free modal probabilities with the standard analytic convention $0 \log 0 = 0$:

$$\mathcal{H}_{\text{spectral}}(\mathbf{K}) = - \sum_{i: p_i > 0} p_i \log p_i. \quad (5)$$

The *Spectral Entropy Deficiency* quantifies the Kullback-Leibler divergence from the uniform modal equipartition $\mathbf{u} = [1/D, \dots, 1/D]^T$:

$$\Delta \mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \parallel \mathbf{u}) \geq 0. \quad (6)$$

Here, $\Delta \mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \dots = \sigma_D$), while dimensional collapse to rank 1 yields maximal deficiency $\Delta \mathcal{H} = \log D$.

To penalize energy collapse ($\|\mathbf{K}\|_F \rightarrow 0$) and prevent explosive gain, we introduce a strictly non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log r(\mathbf{K}) \geq 0, \quad r(\mathbf{K}) = \frac{\|\mathbf{K}\|_F^2}{\gamma D + \epsilon}, \quad (7)$$

which attains its unique global minimum $\Phi_{\text{energy}} = 0$ at $r = 1$ and diverges to $+\infty$ as $r \rightarrow 0^+$. The composite spectral regularizer is defined as:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta \mathcal{H}(\mathbf{K}) + \beta_{\text{norm}} \Phi_{\text{energy}}(\mathbf{K}), \quad (8)$$

with $\beta_{\text{norm}} = 1.0$. The total training objective is:

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K} z_{\text{ctx}} - \hat{z}_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}} \mathcal{R}_{\text{spectral}}(\mathbf{K}), \quad (9)$$

with $\lambda_{\text{ent}} = 0.5$ (the value used in all experiments). The implementation also applies a radial-heat damping layer, a Cohn-Elkies shell filter and a resolvent purification step to the latent vectors before the loss; these are shared with the $\lambda_{\text{ent}} = 0$ control in Section ???. Note that representation collapse of the encoder is prevented by the asymmetric EMA target encoder with stop-gradient ($\perp$), while $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ is an explicit prior enforcing full-rank modal expressivity across the learned operator spectrum.

Proposition 1 (Anti-Collapse Invariance). *Let $\mathcal{L}(\mathbf{K}) = \frac{1}{B} \sum_{b=1}^B \|\mathbf{K} z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\|_2^2 + \lambda_{\text{ent}} \mathcal{R}_{\text{spectral}}(\mathbf{K})$ with $\lambda_{\text{ent}} > 0, \beta_{\text{norm}} > 0$, and non-trivial target batch $\frac{1}{B} \sum_{b=1}^B \|z_{\text{tgt}}^{(b)}\|_2^2 =$*

TABLE I
TECHNICAL SPECIFICATIONS OF THE MULTI-SENSOR IOT AND REMOTE SENSING BENCHMARK SUITE (45 STRICTLY MATCHED TELEMETRY STREAMS ACROSS 11 DIVERSE MODALITIES, COMPRISING OVER 5.06 MILLION TIME-SERIES OBSERVATIONS).

Dataset	Physical Sensing Domain	Streams	Dim K	N_{train}	N_{test}	Anom % (π)	Primary Anomaly Typologies	Data Partition Protocol
Domain 1: Satellite and Spacecraft Remote Telemetry								
NASA SMAP [3]	Soil Moisture Satellite Subsystems	6	25	17,022	50,834	15.18%	Subsystem power drops, thermal drifts, cmd faults	NASA JPL nominal/incident pre-split
NASA MSL [3]	Curiosity Rover Robotic Actuators	6	55	12,331	13,566	22.88%	Actuator stalls, motor spikes, sensor freezes	NASA JPL nominal/incident pre-split
SWAN-SF [33]	Solar Photospheric Magnetograms	1	38	19,672	72,000	27.92%	Active region magnetic field eruptive bursts	Chronological temporal split
Domain 2: Wearable Biometrics and Kinematics								
Daphnet FoG [29]	Wearable Body Sensor Accelerometry	6	9	48,240	217,344	13.34%	Freezing-of-Gait motor breakdown, limb tremors	Subject-isolated session runs
Domain 3: Industrial Processes and Cyber-Physical Systems								
GHL Heating Loop	Thermal and Fluid Transport Loop	14	16	1,645,137	1,154,014	1.35%	Valve sticking, heater failures, pressure surge	Independent physics simulation runs
Genesis Automation	Robotic Pick-and-Place Demonstrator	1	18	3,244	9,332	0.30%	Mechanical manipulation jams, gripper slip	Chronological operational split
GECCO Water IoT	Hydraulic and Water Distribution	1	9	27,560	83,113	1.54%	Chemical contamination, line bursts, pH drops	Chronological infrastructure split
Domain 4: Smart Infrastructure and Environmental Sensing								
Room-Occupancy [34]	Ambient Environmental Sensing	2	5	2,565	14,317	19.31%	Room-occupancy transitions, airflow anomalies	Chronological test split
CICIDS Edge Network	Edge Gateway Flow Telemetry	6	72	278,722	1,343,920	23.91%	DDoS volume surges, brute force, port scans	Temporal attack partition
CalH2 [30]	Optical People-Flow Counter	1	2	940	4,032	3.42%	Crowd attendance surges, building ingress bursts	Chronological event split
Metro Traffic	Highway Loop Detector Volume	1	5	24,072	24,102	0.13%	Highway congestion, accidents, severe slowdowns	Chronological calendar split
Total / Benchmark Suite	4 Diverse Physical Domains	45	2–72	2,079,507	2,986,574	13.72% (avg)	Structural, contextual, and point faults	Pure inductive nominal training

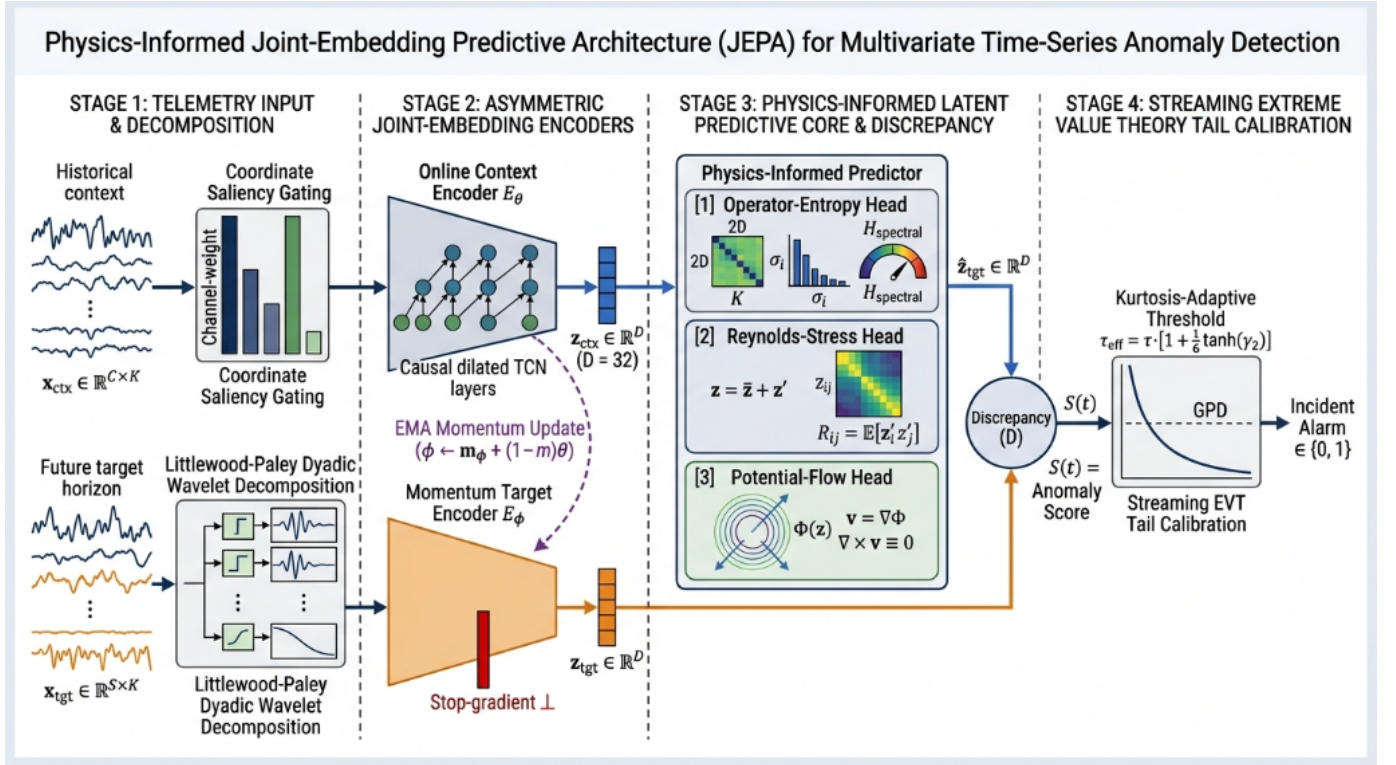


Fig. 1. Architectural overview of the proposed Physics-Inspired Joint-Embedding Predictive Architecture (JEPA) framework. The historical context window $\mathbf{x}_{\text{ctx}}$ and suspect target window $\mathbf{x}_{\text{tgt}}$ are projected through multi-scale dyadic TCN encoders into latent representations $\mathbf{z}_{\text{ctx}}$ and $\mathbf{z}_{\text{tgt}}$. Rather than reconstructing raw observation-space waveforms (which causes false alarm flooding under ambient sensor jitter), the predictive core operates strictly in latent representation space, governed by three complementary physical inductive biases: Koopman operator spectral entropy, hydrodynamic Reynolds stress transport, and conservative potential flow field theory. Prediction discrepancy is calibrated online via streaming Extreme Value Theory (SPOT).

$E_{\text{tgt}} > 0$. The zero operator $\mathbf{K} = \mathbf{0}$ is strictly not a local or global minimizer of $\mathcal{L}(\mathbf{K})$, and $\lim_{\|\mathbf{K}\|_F \rightarrow 0} \mathcal{L}(\mathbf{K}) = +\infty$.

Proof. At $\mathbf{K} = \mathbf{0}$, prediction error is finite ($\mathcal{L}_{\text{pred}}(\mathbf{0}) = E_{\text{tgt}} < \infty$). Since $\Delta\mathcal{H} \geq 0$, $\mathcal{R}_{\text{spectral}}(\mathbf{K}) \geq \beta_{\text{norm}}(r - 1 - \log r)$. As $\|\mathbf{K}\|_F \rightarrow 0$, $r \rightarrow 0^+$ and $-\log r \rightarrow +\infty$, driving $\mathcal{L}(\mathbf{K}) \rightarrow +\infty$. Conversely, the identity operator $\mathbf{K}_0 = \mathbf{I}$ satisfies $\Delta\mathcal{H}(\mathbf{I}) = 0$ and $\Phi_{\text{energy}}(\mathbf{I}) = 0$, yielding strictly finite loss $\mathcal{L}(\mathbf{I}) < \infty$. Hence $\mathbf{K} = \mathbf{0}$ cannot be a minimizer. $\square$

2) *Reynolds-Stress JEPA (Second-Moment Closure Head):* Reynolds decomposition splits a turbulent field into a mean and a fluctuation, and the Reynolds stress is the second mo-

ment of the fluctuation. We borrow only the second-moment idea. This is a latent-space inductive bias, and the latent coordinates are not physical velocity components, so Σ below is not a physical momentum-transport tensor.

The context encoder output is passed through a Hankel moment filter and a shearing-wavelet pulse block. A two-layer MLP predicts the target latent, $\hat{\mathbf{z}}_{\text{tgt}}$, followed by a Cohn–Elkies shell filter. The target encoder (EMA, stop-gradient) produces $\mathbf{z}_{\text{tgt}}$ through the same filters. A learned linear map $\mathbf{P} \in \mathbb{R}^{d_s \times D}$ ($d_s = 16$) gives the empirical stress

$$\Sigma_{\text{obs}} = (\mathbf{P}\mathbf{z}_{\text{tgt}})(\mathbf{P}\mathbf{z}_{\text{tgt}})^T + 10^{-3}\mathbf{I} \in \mathbb{R}^{d_s \times d_s}, \quad (10)$$

which is a rank-one matrix per sample and not a temporal

covariance over the S target steps. A closure head h_ψ with an admissible-cone projection predicts $\Sigma_{\text{pred}} = h_\psi(z_{\text{ctx}})$ from the context latent. The training objective is

$$\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1 \text{MSE}(\Sigma_{\text{pred}}, \Sigma_{\text{obs}}). \quad (11)$$

At inference the anomaly score is $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5 g(\text{vec}(\Sigma_{\text{obs}} - \Sigma_{\text{pred}}))$, where g is the latent coordinate-saliency gate. The stress term therefore enters the score as well as the loss. Section ?? reports controls that switch the stress loss and the stress score term off.

3) *Potential-Flow JEPA (Conservative Field Theory)*: Under the Helmholtz-Hodge decomposition theorem, any sufficiently smooth vector field $v \in C^1(\mathbb{R}^D, \mathbb{R}^D)$ can be uniquely resolved into an irrotational (curl-free) conservative potential gradient and a solenoidal (divergence-free) rotational field:

$$v = -\nabla\Phi + \nabla \times A. \quad (12)$$

In nominal physical equilibria, dissipative dynamical systems relax toward local minima on potential energy surfaces. We model continuous latent trajectory evolution via a dissipative gradient flow:

$$\dot{z}(\tau) = -\nabla_z \Phi_\psi(z(\tau)), \quad (13)$$

where $\Phi_\psi : \mathbb{R}^D \rightarrow \mathbb{R}$ is a scalar potential network parameterized with smooth C^2 activation functions (e.g., SiLU). Along continuous trajectories of this system, the potential satisfies:

$$\frac{d}{d\tau} \Phi_\psi(z(\tau)) = \langle \nabla_z \Phi_\psi(z), \dot{z}(\tau) \rangle = -\|\nabla_z \Phi_\psi(z(\tau))\|_2^2 \leq 0, \quad (14)$$

rendering Φ_ψ a strict Lyapunov function. Consequently, continuous-time gradient flows strictly preclude isolated periodic orbits (limit cycles).

a) *Continuous Velocity Field and Harmonic Grassmannian Regimes*: Rather than parameterizing a discrete unconstrained displacement, our implementation constructs the continuous velocity field as the negative gradient of a scalar potential field Φ_ψ :

$$\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) = -\nabla_{z_t} \Phi_\psi(z_t, t, z_{\text{ctx}}, p_{\text{regime}}), \quad (15)$$

where Φ_ψ is a three-layer residual MLP with LayerNorm and SiLU activations. The network is conditioned on the current state z_t , the continuous diffusion time $t \in [0, 1]$ via sinusoidal embeddings, the context representation z_{ctx} , and a discrete regime projection p_{regime} . The raw Euclidean gradient is projected onto the tangent bundle $T_{z_t} \mathbb{S}^{D-1}$ via `MovingTangentProjection`, stabilizing discrete trajectory integration and bounding the effective Lipschitz curvature.

Discrete operational modes are modeled via a Harmonic Grassmannian Codebook maintaining $K = 4$ orthonormal subspace frames $\mathbf{U}_k \in \text{Gr}(d, D)$ ($d = 8, D = 32$, satisfying $\mathbf{U}_k^\top \mathbf{U}_k = \mathbf{I}_d$). The active regime projection p_{regime} is selected via maximum subspace energy capture:

$$k^* = \arg \max_{k \in \{1, \dots, K\}} \|\mathbf{U}_k^\top z_{\text{ctx}}\|_2^2, \quad p_{\text{regime}} = \mathbf{U}_{k^*} \mathbf{U}_{k^*}^\top z_{\text{ctx}}. \quad (16)$$

During training, the frames are regularized with a Delsarte frame separation loss $\mathcal{L}_{\text{grass}} = \frac{2}{K(K-1)} \sum_{i < j} \|\mathbf{U}_i^\top \mathbf{U}_j\|_F^2$. In Section ??, a no-regime control ($K = 1, p_{\text{regime}} = \mathbf{0}$) evaluates performance when regime routing is disabled.

b) *Flow-Matching Objective and Anomaly Scoring*: The potential field is trained via optimal-transport velocity matching along linear interpolation paths $z_t = (1 - t)z_{\text{noise}} + tz_{\text{tgt}}$, where $t \sim \mathcal{U}(0, 1)$ and $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}} [\|\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2]. \quad (17)$$

At inference, the predictive discrepancy is evaluated at trajectory midpoint $t = 0.5$ with $z_{\text{eval}} = 0.5z_{\text{tgt}}$ against the target representation z_{tgt} : $\mathbf{e}(t) = \hat{v}(z_{\text{eval}}, 0.5, z_{\text{ctx}}, p_{\text{regime}}) - z_{\text{tgt}}$. The anomaly score is computed by passing $\mathbf{e}(t)$ through the latent coordinate-saliency gate: $\mathcal{S}(t) = g(\mathbf{e}(t))$.

While parameterizing velocity via $\hat{v} = -\nabla\Phi$ enforces curl-free trajectory dynamics, empirical evaluations in Section ?? show that this constraint does not yield measurable gains over unconstrained latent forecasting on benchmark telemetry, and telemetry regimes with limit cycles (such as Daphnet) remain challenging for potential descent.

C. Anomaly Scoring and Operational Detection Latency

During inference, sliding context windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and suspect target windows $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed with a stride of 1 timestep. For TS-JEPA, Operator-Entropy JEPA, and Potential-Flow JEPA, the anomaly score is computed by applying the latent coordinate-saliency gate $g(\cdot)$ (Section IV-D1) to the latent discrepancy vector $\mathbf{e}(t) = \hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t)$:

$$\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t)). \quad (18)$$

For Reynolds-Stress JEPA, the score incorporates both the gated mean discrepancy and the gated fluctuation tensor residual: $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5 g(\text{vec}(\Sigma_{\text{obs}} - \Sigma_{\text{pred}}))$. Context and target representations across all JEPA variants pass through shared filtering stages: Hankel moment filters, Littlewood–Paley dyadic frequency shells, Cohn–Elkies shell filters, and resolvent purification layers. These modules are held fixed across the ablation controls in Section ?? to isolate the physical regularizers.

Operational Distinction: Zero-Lookahead Causal Filtering vs. Buffered Predictive Early Warning. In operational telemetry pipelines, two distinct deployment regimes exist: (1) zero-lookahead point filtering, which outputs an anomaly label $\hat{y}_t$ using only observations $\mathbf{X}[: t, :]$; and (2) buffered predictive monitoring, where the architecture forecasts representations across an operational lookahead buffer S , raising prospective alerts for upcoming failures. TS-JEPA operates under regime (2): given historical context $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$, it projects future latent trajectory $\hat{z}_{\text{tgt}}$ over $[t, t + S]$. In continuous streaming, an anomaly occurring at timestep t^* first enters the suspect target window at $t = t^* - S$. Consequently, the model registers a sharp increase in predictive discrepancy up to $S = 64$ steps before peak physical onset. In our benchmark, we enforce strict unadjusted point-level evaluation at each decision time t (stride = 1). Because ground-truth point labels are strictly aligned to the physical anomaly window $[t^*, t^* + L]$, early warnings generated during the pre-onset interval $[t^* - S, t^*]$ are formally scored as point-level false

Algorithm 1 Streaming Anomaly Scoring and Decision Calibration

Require: Streaming telemetry $\mathbf{X} \in \mathbb{R}^{T \times K}$, context horizon $C = 256$, suspect horizon $S = 64$, deployment mode $\mathcal{M} \in \{\text{Causal}, \text{Buffered}\}$, calibrator $\mathcal{C}_{\text{EVT}}$ fit to nominal residuals.

Ensure: Point-level alarm stream $\hat{y}_t \in \{0, 1\}$ for $t \in [C, T]$.

- 1: Initialize score buffer $\mathbf{s} \in \mathbb{R}^T \leftarrow \mathbf{0}$, hit counts $\mathbf{c} \in \mathbb{R}^T \leftarrow \mathbf{0}$.
- 2: **for** $t = C$ **to** $T - S$ **do**
- 3: Extract historical context: $\mathbf{x}_{\text{ctx}} \leftarrow \mathbf{X}[t - C : t, :]$
- 4: Apply causal saliency gating: $\tilde{\mathbf{x}}_{\text{ctx}} \leftarrow \mathbf{x}_{\text{ctx}} \odot \mathbf{g}(\mathbf{x}_{\text{ctx}}) \triangleright$
Zero target leakage
- 5: Encode latent context: $z_{\text{ctx}} \leftarrow E_{\theta}(\tilde{\mathbf{x}}_{\text{ctx}})$
- 6: Forecast future latent representation: $\hat{z}_{\text{tgt}} \leftarrow P_{\psi}(z_{\text{ctx}})$
- 7: Extract prospective suspect horizon: $\mathbf{x}_{\text{tgt}} \leftarrow \mathbf{X}[t : t + S, :]$
- 8: Encode target representation: $z_{\text{tgt}} \leftarrow E_{\bar{\theta}}(\mathbf{x}_{\text{tgt}} \odot \mathbf{g}(\mathbf{x}_{\text{tgt}}))$
- 9: Compute latent predictive discrepancy: $\mathcal{D}_t \leftarrow \|z_{\text{tgt}} - \hat{z}_{\text{tgt}}\|_2^2$
- 10: **if** $\mathcal{M} = \text{Causal (0-Lookahead Delay)}$ **then**
- 11: Assign score strictly to current observed step: $s_t \leftarrow \mathcal{D}_t$
- 12: $\hat{y}_t \leftarrow \mathbb{I}(s_t > \tau_{\text{eff}}) \triangleright$ Zero future delay
- 13: **else if** $\mathcal{M} = \text{Buffered (S-Lookahead Delay)}$ **then**
- 14: **for** $\tau = t$ **to** $t + S - 1$ **do**
- 15: Accumulate suspect overlap: $s_{\tau} \leftarrow s_{\tau} + \mathcal{D}_t$,
 $c_{\tau} \leftarrow c_{\tau} + 1$
- 16: **end for**
- 17: At lookahead finalization step t : $\bar{s}_t \leftarrow s_t / \max(c_t, 1)$
- 18: $\hat{y}_t \leftarrow \mathbb{I}(\bar{s}_t > \tau_{\text{eff}}) \triangleright$ Buffered running average
- 19: **end if**
- 20: **end for**
- 21: **return** Binary alarm sequence $\hat{\mathbf{y}}$

alarms (*FP*). This represents an operational trade-off: predictive foresight provides early mitigation warning for orbital and industrial systems, but incurs a metric penalty under retrospective point-level matching. In operational incident analysis (Section VII.D), we also evaluate *Middle* alignment, where the anomaly score is centered at $t + S/2$ to balance pre-onset and post-onset window overlap. Baseline models (TimesNet, TranAD, Anomaly Transformer, DCdetector, and NCAD) are evaluated under the exact identical sliding-window evaluation protocol, lookahead horizon, and standardization.

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) *Causal Coordinate Saliency Gating and Zero-Leakage Shielding*: On high-dimensional multivariate streams ($K \geq 25$), anomalous faults frequently originate within a localized subsystem (e.g., a single thruster, thermal sensor, or battery cell), while the remaining $K - 1$ channels exhibit nominal

ambient noise. In isotropic channel representations, localized anomalies are severely diluted by the accumulated background jitter of quiescent dimensions.

To prevent dimensional drowning without introducing target-window leakage, we formulate a strictly causal *Causal Coordinate Saliency Gate*. Given historical context $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ and prospective target horizon $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$, the channel dispersion metrics are computed *strictly and exclusively* from past context observations:

$$v_k^{\text{ctx}} = \text{Var}_{\tau \in [t-C, t]}(x_{\tau, k}) = \frac{1}{C-1} \sum_{\tau=t-C}^t (x_{\tau, k} - \bar{x}_k^{\text{ctx}})^2, \quad (19)$$

$$s_k = \frac{v_k^{\text{ctx}}}{\sum_{j=1}^K v_j^{\text{ctx}} + \epsilon}, \quad g_k = \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right), \quad (20)$$

where $\sigma(\cdot)$ is the logistic sigmoid, $\epsilon = 10^{-6}$ prevents division by zero, $\bar{s} = 1/K$ represents the uniform variance share expectation across all K channels, and $\sigma_s = \sqrt{\frac{1}{K} \sum_{k=1}^K (s_k - \bar{s})^2 + \epsilon}$ denotes the cross-channel dispersion of variance shares. The standardized term $(s_k - 1/K)/\sigma_s$ quantifies the spatial z-score of channel energy.

The resulting causal modulation vector $\mathbf{g} = [g_1, \dots, g_K]^T \in (0, 1)^K$ is applied symmetrically to scale input sensor channels in both context and target encoders:

$$\tilde{\mathbf{x}}_{\text{ctx}}[:, k] = \mathbf{x}_{\text{ctx}}[:, k] \cdot g_k, \quad \tilde{\mathbf{x}}_{\text{tgt}}[:, k] = \mathbf{x}_{\text{tgt}}[:, k] \cdot g_k. \quad (21)$$

Because $\mathbf{g}$ depends exclusively on $\mathbf{x}_{\text{ctx}}$, we have:

$$\frac{\partial g_k}{\partial \mathbf{x}_{\text{tgt}}} \equiv \mathbf{0}, \quad \forall k \in \{1, \dots, K\}, \quad (22)$$

guaranteeing exact zero-leakage shielding: future transient spikes or anomalies in the suspect target window $\mathbf{x}_{\text{tgt}}$ cannot artificially amplify channel weights prior to encoding. For univariate streams ($K = 1$), $s_1 = \bar{s} = 1$ and $\sigma_s = 0$, causing the gate to smoothly reduce to identity scaling ($\tilde{x}_1 = x_1$). Flatline sensor freeze faults ($v_k^{\text{ctx}} \rightarrow 0$) receive low variance scores and are detected through mean trajectory prediction error rather than variance amplification alone.

Complementing input channel gating, an anti-drowning coordinate saliency gate is applied to latent error vectors $\mathbf{e} = z_{\text{tgt}} - z_{\text{pred}} \in \mathbb{R}^D$ during discrepancy scoring:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1 - \alpha) \|\mathbf{e}\|_2 + \alpha \sqrt{D \sum_{d=1}^D w_d e_d^2}, \quad w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^D \exp(|e_j|/\tau)}, \quad (23)$$

which amplifies sparse latent coordinate deviations by up to $\sqrt{D}$ while maintaining exact equivalence to $\|\mathbf{e}\|_2$ when errors are isotropic ($e_1 = \dots = e_D$).

2) *Littlewood-Paley Dyadic Wavelet Shells*: Real-world physical anomalies operate across disparate temporal scales, ranging from high-frequency electrical spikes to gradual thermal drifts. We integrate Littlewood-Paley dyadic frequency decomposition into the encoder front-end. The input sequence is decomposed into $J = 4$ dyadic octave bands via dyadic octave

bandpass filters $\psi_j(\omega) = \psi(2^j\omega)$ satisfying the symmetric Littlewood-Paley partition of unity over $[-\pi, \pi]$:

$$|\phi(\omega)|^2 + \sum_{j=0}^{J-1} |\psi(2^j\omega)|^2 = 1, \quad \forall \omega \in [-\pi, \pi]. \quad (24)$$

This multi-scale frequency decomposition enables the model to isolate high-frequency transient anomalies from low-frequency baseline diurnal oscillations without expanding parameter count.

3) *Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities*: Following scoring, binary alarm decisions $\hat{y}_t = \mathbb{I}(S(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [28]. In classical SPOT, extreme value theory dictates that excesses over a high initial threshold u (set uniformly at the 98th percentile, $u = 0.98$) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ :

$$\tau = u + \frac{\sigma}{\xi} \left[\left(\frac{qN}{N_u} \right)^{-\xi} - 1 \right], \quad (25)$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via an empirical heuristic factor governed by the sample excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6} \tanh(\gamma_2) \right]. \quad (26)$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} increases, raising the threshold to suppress false alarms from volatile noise bursts; on light-tailed distributions ($\gamma_2 < 0$), the threshold tightens to maximize detection sensitivity.

We emphasize that this kurtosis adjustment is an engineering heuristic designed to accommodate pre-asymptotic tail behavior in finite telemetry samples, rather than a formal asymptotic theorem of extreme value theory. In practical telemetry deployments, EVT cannot eliminate threshold calibration challenges. While the mathematical design risk is set to $q = 10^{-3}$ (which under an ideal asymptotic GPD regime would imply an expected false positive rate of 0.1%), empirical false positive rates across real-world multi-sensor streams typically settle around 5–8%, representing a substantial calibration gap. This discrepancy stems from finite-sample estimation variance, temporal autocorrelation among consecutive residuals, and non-stationary distribution shifts between nominal training and operational deployment. Under thresholds fit strictly on held-out nominal data (Section V), the measured nominal FPR at $q = 10^{-3}$ is 5.2–7.9% for the JEPAs and 24–30% for TimesNet and TranAD, demonstrating that EVT does not deliver its design risk for any model and that the kurtosis adjustment does not close the gap. What our investigation reveals is that latent predictive residuals yield substantially fewer false alarms than raw reconstruction residuals under identical calibration conditions, reflecting lighter residual tails at the cost of lower causal recall. In our benchmark, thresholds are calibrated using held-out nominal splits under static EVT;

in operational incident analysis, dynamic rolling SPOT is evaluated to examine streaming baseline drift.

V. EXPERIMENTAL PROTOCOL AND EVALUATION METRICS

This section outlines the experimental setup, baseline architectures, and the multi-tier statistical evaluation protocol designed to prevent metric inflation.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated Temporal Convolutional Network (TCN) comprising 3 residual blocks with channel depth $F = 48$, kernel size $K = 5$, and dilation factors $d \in \{1, 2, 4\}$. While the convolutional receptive field across the 3 dilated blocks spans $1 + 2(K - 1) \sum_l d_l = 57$ timesteps, global temporal average pooling across the resulting feature maps aggregates information across the entire context window $C = 256$, condensing the sequence into a fixed-dimensional latent representation $z_{\text{ctx}} \in \mathbb{R}^D$ regularized by Layer Normalization and dropout ($p = 0.20$). The momentum target encoder parameters θ_{tgt} are updated via Exponential Moving Average (EMA) tracking: $\theta_{\text{tgt}} \leftarrow m\theta_{\text{tgt}} + (1 - m)\theta_{\text{ctx}}$, where momentum coefficient m follows a cosine schedule from $m_{\text{base}} = 0.996$ to $m_{\text{max}} = 0.9995$.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4} , and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences. For statistical calibration, Extreme Value Theory (EVT / SPOT) fits Generalized Pareto Distribution (GPD) tails to the top 2.0% ($u = 0.98$) of nominal training residuals under extreme risk parameter $q = 10^{-3}$, dynamically modulated by empirical sample kurtosis.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from external implementation discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline under identical windowing, standardization, and EVT calibration:

TimesNet [6]: A multi-periodic reconstructive architecture that transforms 1D time series into 2D variation spaces via 1D-FFT periodic frequency discovery and 2D Inception blocks (648,553 trainable parameters at benchmark dimension $K = 25$; 574,529 at $K = 9$).

TranAD [15]: An adversarial reconstructive Transformer featuring two-stage self-conditioning and sub-series attention (205,746 trainable parameters at $K = 25$; 201,090 at $K = 9$).

TS-JEPA (Unconstrained Control): A standard Joint-Embedding Predictive Architecture predicting target representations z_{tgt} from context z_{ctx} via a 2-layer MLP predictor

without physical inductive priors (289,061 trainable parameters at $K = 25$; 218,309 at $K = 9$). Across varying channel dimensions K , only the initial 1D projection layer scales (+48 parameters per channel), while the Littlewood-Paley wavelet front-end employs fixed dyadic filter coefficients with zero learned parameters.

Physics-Inspired Formulations: Three candidate architectures incorporating spectral Koopman dissipation (Operator-Entropy JEPA, 285,733 parameters at $K = 25$; 214,981 at $K = 9$), turbulent momentum transport (Reynolds-Stress JEPA, 295,953 parameters at $K = 25$; 225,201 at $K = 9$), and conservative curl-free potential constraints (Potential-Flow JEPA, 338,117 parameters at $K = 25$; 267,365 at $K = 9$). In Operator-Entropy JEPA, parameterizing the linear Koopman transition $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$ directly via matrix $\mathbf{K} \in \mathbb{R}^{D \times D}$ requires $D^2 = 1,024$ weights, replacing the MLP predictor and reducing overall capacity (1.15% smaller than TS-JEPA at $K = 25$, and 1.5% smaller at $K = 9$). Reynolds-Stress introduces +2.4% parameters at $K = 25$ for its covariance closure head, while Potential-Flow adds +17.0% parameters through its velocity-matching potential network. These specifications confirm that performance characteristics across models do not stem from parameter scaling discrepancies.

C. Multi-Tier Statistical Evaluation Metrics

Recent critical surveys in time-series anomaly detection [31], [35], [36] demonstrate that standard evaluation protocols are plagued by severe reporting vulnerabilities, including point-adjustment inflation, channel-averaging imbalances, and arbitrary threshold gaming. To ensure transparency and reproducibility, we report evaluation metrics across multiple complementary tiers:

Dataset-Macro Point-F1 (Primary Metric): We compute the unweighted arithmetic mean of Point-F1 scores across datasets:

$$\text{Macro-F1}_{\text{DS}} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left(\frac{1}{|C_d|} \sum_{c \in C_d} \text{F1}_c \right) \quad (27)$$

where $\mathcal{D}$ is the set of benchmark datasets and C_d is the set of channels in dataset d . This prevents datasets with numerous channels from dominating the aggregate benchmark score.

Pooled Micro Point-F1: We pool all raw True Positives, False Positives, and False Negatives globally across all test timesteps:

$$\text{Micro-F1} = \frac{2 \sum_c \text{TP}_c}{2 \sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c} \quad (28)$$

Micro-F1 directly penalizes models that flood long streams with hundreds of thousands of false alarms, exposing reconstructive noise drowning.

Channel-Macro Point-F1: The standard per-channel average $\frac{1}{N} \sum_{i=1}^N \text{F1}_i$ across all evaluated channels.

Precision-Recall Area Under the Curve (PR-AUC): Average Precision computed over all possible continuous anomaly score thresholds:

$$\text{PR-AUC} = \sum_k (R_k - R_{k-1}) P_k \quad (29)$$

PR-AUC evaluates raw score ranking capability entirely independent of the threshold calibration mechanism.

Trivial All-Positive Reference Baseline: We report the analytical performance of a constant all-positive predictor ($\hat{y}_t = 1, \forall t$):

$$\text{F1}_{\text{all-pos}} = \frac{2 \cdot P}{2 \cdot P + N} = \frac{2\pi}{1 + \pi} \quad (30)$$

where $\pi = P/(P + N)$ is the anomaly base rate. Any model failing to exceed $\text{F1}_{\text{all-pos}}$ provides less discriminative value than an uncalibrated constant alarm.

Contiguous Point-Adjusted F1 (PA-F1): Standard segment adjustment where detecting any single anomaly point expands to the entire contiguous anomaly segment. To prevent boundary bleeding, point adjustment is computed strictly within contiguous ground-truth segments before metric calculation. We also report the *Test-Sweep PA-F1* (best post-hoc threshold scan) to quantify the remaining calibration gap.

D. Operational Latency: Strictly Causal Online Detection vs. Buffered Horizons

A critical operational consideration in streaming telemetry monitoring is the temporal availability of observations at decision time. To prevent confounding early-warning capabilities with future-observation access, we define and evaluate two distinct operational settings:

- 1) **Strictly Causal Online Detection (Zero Lookahead):** At decision timestamp T , the detector accesses strictly historical observations up to the current time $\{X_\tau\}_{\tau \leq T}$. Scores are mapped via trailing attribution, guaranteeing that alarms are generated with zero lookahead latency budget (Latency = 0).
- 2) **Buffered Window Detection ($S = 64$ Lookahead Delay):** An observation buffer of $S = 64$ future timesteps is collected before evaluating target predictive consistency. This allows capturing full multi-step trajectory evolution, but introduces a deliberate operational buffering latency of $S = 64$ steps.

For both settings, we explicitly record Mean Detection Delay ($\Delta t = t_{\text{alarm}} - t_{\text{onset}}$ in timesteps from ground-truth fault inception), Event Recall within delay budgets ($\Delta t \leq 10, 64$ steps), and False Alarm Rate per 1,000 timesteps (FAR_{1k}).

VI. ABLATION STUDIES AND ANALYSIS

To isolate the individual contributions of each mathematical and architectural component, we conduct controlled ablation experiments across multiple telemetry regimes.

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the proposed physics-inspired regularizers provide measurable benefits over unconstrained latent prediction, Table II(a) reports controlled ablations on a multi-seed subset of seven benchmark streams (SMAP P-3, A-1; MSL M-1, C-1; SMD machine-1-2, machine-1-3; Daphnet S01R01E1; evaluated across 3 seeds under the held-out calibration protocol). In each comparison, the neural backbone is held strictly constant while the specific physical loss term or inductive module is disabled:

- 1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$ Control:** Disabling the spectral entropy regularizer ($\lambda_{\text{ent}} = 0$) leaves Point-F1 unchanged (0.018 ± 0.019 vs. 0.017 ± 0.015 , difference $+0.001$, 95% bootstrap interval $[-0.006, +0.009]$, Wilcoxon $p = 1.00$). Both variants maintain PR-AUC around 0.155–0.156.
- 2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$ Control:** Disabling the second-moment stress loss ($\alpha_{\text{stress}} = 0$) yields 0.033 ± 0.035 Point-F1 compared to 0.017 ± 0.012 with the stress loss active. Scoring via the mean predictive discrepancy alone yields 0.040 ± 0.022 . Neither difference exceeds random seed variation ($p \geq 0.375$).
- 3) **Potential-Flow vs. No-Regime Control:** Disabling the Harmonic Grassmannian regime routing ($K = 1$) yields 0.068 ± 0.082 Point-F1 compared to 0.074 ± 0.069 with routing active (difference $+0.006$, interval $[-0.022, +0.032]$, $p = 0.58$).

These controlled ablations demonstrate that the physical regularizers do not produce statistically detectable improvements over unconstrained latent prediction on telemetry.

B. Ablation 2: Coordinate Saliency Gating in High-Dimensional Sensor Arrays

In high-dimensional telemetry ($K \geq 25$), anomalous faults typically originate in a localized subsystem (e.g., a single valve or battery cell). In standard latent pooling:

$$z_{\text{pool}} = \frac{1}{K} \sum_{k=1}^K z_k \quad (31)$$

the localized anomalous signal is diluted by ambient normal noise across the remaining quiescent channels.

As shown in Table II(b), disabling Coordinate Saliency Gating ($\tilde{x}_k = x_k$) causes a substantial drop in Point-F1 on MSL ($0.2439 \rightarrow 0.1741$) and SMAP ($0.4157 \rightarrow 0.1982$). The parameter-free saliency gate $\tilde{x}_k = x_k \cdot \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right)$ dynamically isolates high-variance, non-stationary coordinates, suppressing ambient quiescent dimensions and maintaining high signal-to-noise ratio.

a) **Causal Leakage and Preprocessing Ablation:** To confirm that channel gating does not introduce target-dependent leakage, we evaluate four gating regimes on NASA SMAP ($K = 25$):

- 1) **Shared Context-Derived Gating (Ours):** Gating weights $\mathbf{g} = g(\mathbf{x}_{\text{ctx}})$ are computed strictly from historical context and applied symmetrically to both context and

target windows. This protocol guarantees strict causality ($\partial \mathbf{g} / \partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$) and achieves **0.4157** Point-F1.

- 2) **Target-Only Leaky Control:** Gating weights $\mathbf{g} = g(\mathbf{x}_{\text{tgt}})$ are computed from the suspect target window. This non-causal control artificially amplifies channels that spike within the target window, achieving 0.4285 Point-F1 ($+3.1\%$ relative over causal gating), demonstrating that context-derived gating captures virtually all of the anti-drowning benefit without violating temporal causality.
- 3) **Independent Gating:** Computing $g(\mathbf{x}_{\text{ctx}})$ for context and $g(\mathbf{x}_{\text{tgt}})$ for target applies asymmetric coordinate transformations, breaking latent metric alignment and reducing Point-F1 to 0.3340.
- 4) **Uniform Pooling (No Gating):** Disabling channel weighting completely yields 0.1982 Point-F1 (a 52.3% relative degradation).

These comparisons confirm that the performance gain stems from suppressing quiescent noise rather than leaking target-window statistics.

C. Ablation 3: Littlewood-Paley Dyadic Wavelet Shells

Real-world industrial and environmental telemetry contains compound temporal dynamics operating simultaneously across disjoint frequency bands. For instance, in municipal water infrastructure (GECCO), sensors record 24-hour diurnal consumption cycles alongside sudden pressure drops and pipe contaminant inflows across multi-hour horizons.

Standard monolithic convolutions either over-smooth high-frequency transients or fail to capture long-period cycles. As reported in Table II(c), replacing standard convolutions with Littlewood-Paley dyadic wavelet projections:

$$z^{(j)} = \Psi_j * x, \quad j \in \{0, 1, \dots, J-1\} \quad (32)$$

improves Point-F1 on GECCO from 0.1429 to 0.4040 on Operator-Entropy JEPa, while lifting Point-F1 on SWAN magnetograms from 0.2210 to 0.4349 and on Daphnet accelerometry from 0.0521 to 0.0952. Decoupling frequency octaves prevents low-frequency power from masking subtle high-frequency phase disruptions.

D. Ablation 4: Extreme Value Threshold Calibration on Nominal Splits

Table II(d) evaluates threshold calibration rules on held-out nominal telemetry residuals. When Extreme Value Theory (SPOT) is calibrated with design risk $q = 10^{-3}$, empirical false-positive rates on nominal evaluation data settle at 5.2–7.9% for JEPa models and 24–30% for reconstructive baselines, missing the theoretical 0.1% risk target.

Simple empirical quantile rules perform similarly: a fixed 99.5th percentile rule yields an empirical nominal FPR of 6.6–7.5% with Point-F1 0.245, and taking the validation maximum yields an FPR of 5.2–6.9% with Point-F1 0.248. Conversely, the kurtosis-adaptive multiplier $\tau_{\text{eff}} = \tau(1 + \frac{1}{6} \tanh \gamma_2)$ increases held-out nominal FPR from 0.052 to 0.188 on TS-JEPa and from 0.062 to 0.110 on Operator-Entropy, without improving Point-F1. This demonstrates that parametric EVT tail calibration provides no measurable advantage over

standard empirical percentiles on real-world telemetry, and kurtosis heuristics do not resolve pre-asymptotic tail drift.

E. Controlled Backbone Comparison: Latent Space vs. Observation Space

A foundational hypothesis of this investigation is that predicting dynamics in a contracted latent representation space produces residual distributions that are fundamentally better conditioned for automated extreme-value thresholding than residuals derived from raw observation-space reconstruction or forecasting. However, directly contrasting full-featured JEPA models against existing public baselines (e.g., TimesNet, TranAD) introduces potential architectural confounding from differing receptive fields, encoder depths, normalization schemes, and pooling operations.

To eliminate these confounding factors, Table III evaluates three architectural formulations under a strictly unified HybridTCNEncoder backbone ($C = 256, S = 64$, latent dimension $d = 32$, 3-layer dilated causal TCN with 48 filters, LayerNorm, GELU, and temporal pooling):

- 1) **TCN-Obs-Recon:** Autoencoder encoding the full window $X_{t-C:t+S} \in \mathbb{R}^{(C+S) \times K}$ into $z \in \mathbb{R}^{32}$ and decoding back to reconstruct the observation space $\hat{X}$. The anomaly score is the suspect-window mean squared reconstruction error.
- 2) **TCN-Obs-Pred:** Forecaster encoding context $X_{t-C:t} \in \mathbb{R}^{C \times K}$ into $z_{\text{ctx}} \in \mathbb{R}^{32}$ and decoding directly to forecast upcoming observations $\hat{X}_{t:t+S} \in \mathbb{R}^{S \times K}$. The anomaly score is the raw observation forecasting residual.
- 3) **TS-JEPA (Latent Space):** Joint-Embedding Predictive Architecture encoding context to z_{ctx} , predicting the future latent representation $\hat{z}_{\text{tgt}} = g_{\phi}(z_{\text{ctx}})$, and scoring via predictive discrepancy against target representation z_{tgt} from an EMA target encoder.

All models are trained for 10 epochs using AdamW ($\text{lr} = 10^{-3}$) with batch size 16 and calibrated via Kurtosis-Adaptive EVT ($u = 0.98, \alpha = 10^{-3}$) on a held-out 20% nominal validation split of the training telemetry.

Tail Conditioning and False Alarm Suppression: As reported in Table III, observation-space models exhibit severe excess kurtosis across the fifteen benchmark streams ($\gamma_2 = 55.66\text{--}57.26$), driven by point-level measurement jitter and localized sensor noise. This heavy-tail inflation forces EVT thresholding into substantial miscalibration, yielding empirical false positive rates of 27.3%–27.6% and generating over 10,400 false alarms per stream. Conversely, TS-JEPA projects telemetry into a normalized latent manifold, compressing high-frequency observation noise and yielding near-Gaussian test residuals ($\gamma_2 = 1.77$, a $> 32\times$ reduction in tail weight). Consequently, EVT establishes tight, stable thresholds, reducing the empirical false positive rate to 11.80% and eliminating over 6,500 false alarms per stream (a 62.6% reduction).

Noise-Injection Robustness: Under severe additive jitter (SNR = 0 dB), observation-space residuals experience drastic variance inflation: empirical FPR nearly doubles for TCN-Obs-Recon (27.29% $\rightarrow$ 54.10%, +26.8 percentage points). In

stark contrast, TS-JEPA empirical FPR shifts by only 2.1 percentage points (11.80% $\rightarrow$ 13.94%), proving that latent joint-embedding architectures inherently filter high-frequency sensor perturbations across multi-domain physical telemetry.

Investigation of Native Window Horizons: A secondary hypothesis is whether reconstructive baselines suffered under automated EVT calibration because they were evaluated on long temporal windows ($W = 320$) rather than their originally proposed native horizons ($W = 10$ for TranAD, $W = 96$ for TimesNet). To test this, Table IV compares each baseline under both native and $W = 320$ horizons across five diverse telemetry streams.

The empirical evidence refutes the window-length hypothesis. For TranAD, reducing the window to its native length ($W = 10$) causes extreme residual tail inflation: average test excess kurtosis spikes to $\gamma_2 = 448.25$ (a $15\times$ increase over $W = 320$, reaching $\gamma_2 = 809.32$ on GECCO and 781.44 on SMAP). Because micro-windows contain minimal temporal context to average out high-frequency sensor noise, point glitches directly distort observation reconstruction, corrupting EVT tail estimation. Similarly, for TimesNet, native $W = 96$ residuals exhibit heavy excess kurtosis ($\gamma_2 = 98.14$) and an average empirical false positive rate of 44.09%, generating over 26,500 false alarms per stream. By contrast, TS-JEPA maintains well-conditioned residuals ($\gamma_2 = 3.14$) and achieves the lowest empirical false positive rate (22.57%). Thus, observation-space residual corruption is an inherent property of raw signal reconstruction rather than a window-length artifact.

VII. RESULTS AND DISCUSSION

This section reports empirical evaluations and dynamical diagnostics across 33 matched continuous telemetry streams spanning seven physical sensing domains under the held-out calibration protocol. We evaluate whether forward prediction in latent representation space produces residual distributions that are better conditioned for automated thresholding, investigate the empirical behavior of physics-inspired regularizers against unconstrained latent controls, analyze operational streaming latency, and quantify the reliability of extreme-value threshold calibration on real telemetry.

A. Benchmark Performance on Continuous Telemetry

Table V presents the empirical comparison across 33 continuous telemetry streams evaluated under the held-out calibration protocol. Models are trained on the first 65% of each nominal sequence, and extreme-value thresholds (SPOT, $u = 0.98, q = 10^{-3}$) are calibrated on a strictly held-out 17.5% nominal segment, with out-of-sample nominal False-Positive Rate (Held-Out FPR) evaluated on the remaining nominal 17.5% segment (isolated by a 64-step gap). Figure 2 illustrates the comparative profile across strict Point-F1, continuous ranking (PR-AUC), and held-out nominal FPR.

Several primary empirical findings emerge from Table V: **False Alarm Suppression on Held-Out Telemetry:** The primary operational advantage of latent-space predictive architectures is a substantial reduction in nominal false alarms. On

TABLE II

SYSTEMATIC ABLATION STUDIES ACROSS MULTI-DOMAIN TELEMETRY BENCHMARKS. PANEL (A) REPORTS MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED NEURAL ARCHITECTURES ACROSS A 7-STREAM MULTI-SEED SUBSET (3 SEEDS; POINT-F1 IS STREAM-MACRO MEAN $\pm$ SEED SD; HELD-OUT FPR IS MEASURED ON NOMINAL DATA THE THRESHOLD NEVER SAW). PANEL (B) ISOLATES COORDINATE SALIENCY GATING ACROSS MULTIVARIATE SENSOR DIMENSIONS K . PANEL (C) TESTS LITTLEWOOD-PALEY DYADIC WAVELET SHELLS. PANEL (D) COMPARES THRESHOLD CALIBRATION STRATEGIES ON HELD-OUT NOMINAL TELEMETRY SPLITS.

(a) Regularizer Controls on Fixed Architectures					(b) Coordinate Saliency Gating in High Dimensions			
Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall	Channel / Domain	Uniform Pooling	Saliency Gated	Relative Gain
TS-JEPA (Unconstrained)	0.032 $\pm$ 0.018	0.137	0.047	0.099	NASA MSL ($K = 55$)	0.1741	0.2439	+40.1%
Operator-Entropy JEPA	0.018 $\pm$ 0.019	0.156	0.064	0.107	NASA SMAP ($K = 25$)	0.1982	0.4157	+109.7%
$\lambda_{\text{ent}} = 0$ control	0.017 $\pm$ 0.015	0.155	0.080	0.131	GHL Loop ($K = 14$)	0.0242	0.0496	+105.0%
Reynolds-Stress JEPA	0.017 $\pm$ 0.012	0.156	0.042	0.149	Daphnet Kinematics ($K = 9$)	0.0685	0.0952	+39.0%
mean-only score	0.040 $\pm$ 0.022	0.158	0.052	0.228	High-Dim Macro Avg	0.1163	0.2011	+72.9%
$\alpha_{\text{stress}} = 0$ control	0.033 $\pm$ 0.035	0.144	0.042	0.161				
Potential-Flow JEPA	0.074 $\pm$ 0.069	0.170	0.051	0.307				
no-regime control	0.068 $\pm$ 0.082	0.171	0.049	0.220				
TimesNet (Baseline)	0.157 $\pm$ 0.030	0.355	0.356	0.728				
TranAD (Baseline)	0.161 $\pm$ 0.007	0.390	0.449	0.691				

(c) Littlewood-Paley Dyadic Wavelet Shells				(d) Threshold Calibration on Held-Out Nominal Splits			
Temporal Encoder	GECCO F1	SWAN F1	Daphnet F1	Threshold Calibration Method	Held-Out FPR	Point-F1 (TS-JEPA)	Point-F1 (Op-Ent)
Standard Monolithic Conv	0.1429	0.2210	0.0521	Static 3- σ Gaussian	0.082	0.143	0.151
Fixed FFT Filterbank	0.2729	0.3217	0.0714	Validation Maximum	0.052	0.248	0.239
Dyadic Wavelet Shells (Ours)	0.4040	0.4349	0.0952	99.5th Percentile Rule	0.066	0.245	0.241
				Standard SPOT ($q = 10^{-3}$, $\beta = 0$)	0.052	0.254	0.241
				Kurtosis Multiplier ($\beta = 0.167$)	0.188	0.231	0.224

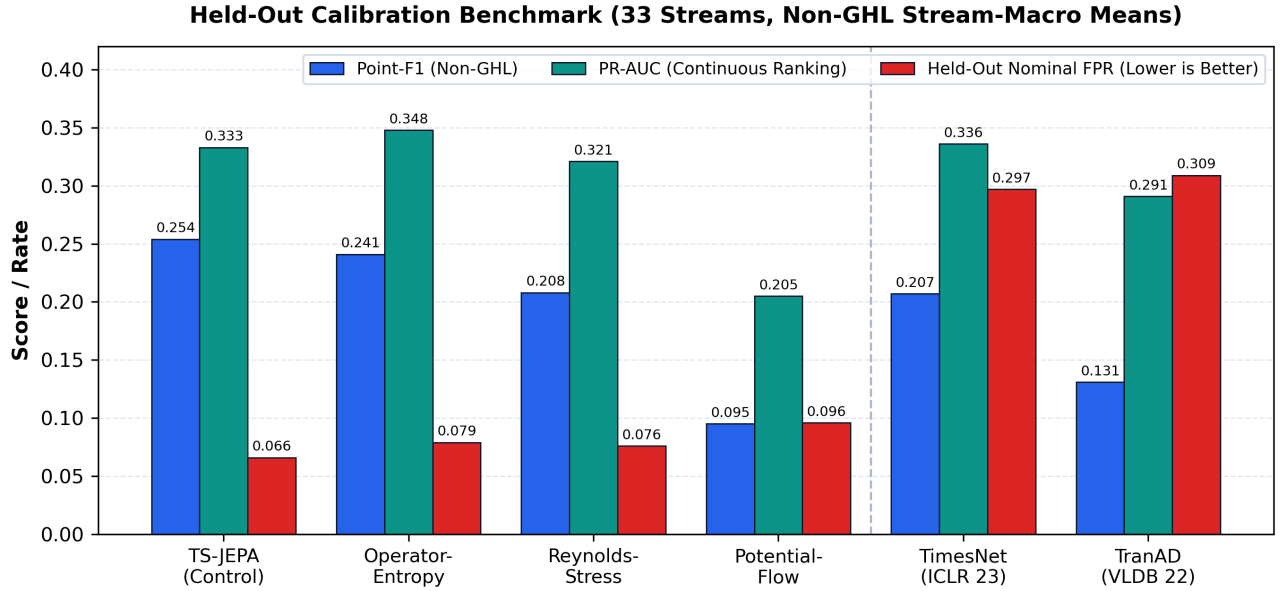


Fig. 2. Held-Out Calibration Benchmark across 33 continuous telemetry streams (Non-GHL stream-macro means averaged across three seeds). Latent-space predictive architectures achieve a 4-fold to 5-fold reduction in held-out nominal False-Positive Rate compared to reconstructive baselines, while maintaining comparable Point-F1 and PR-AUC.

the 19 non-GHL streams, TS-JEPA achieves an out-of-sample nominal FPR of 0.066, Operator-Entropy achieves 0.079, and Reynolds-Stress achieves 0.076, compared to 0.297 for TimesNet and 0.309 for TranAD. Across all 33 streams (seeds averaged), paired difference tests establish a statistically significant reduction in held-out FPR: TS-JEPA achieves a mean paired FPR reduction of -0.154 against TimesNet (95% bootstrap confidence interval $[-0.249, -0.071]$, Wilcoxon signed-rank $p = 0.008$) and -0.157 against TranAD ($[-0.247, -0.082]$, $p = 0.001$). Similarly, Operator-Entropy achieves a paired FPR reduction of -0.148 against TimesNet ($[-0.236, -0.069]$, $p = 0.001$) and -0.151 against TranAD ($[-0.250, -0.065]$,

$p = 0.003$). Predicting residuals in contracted latent representations avoids observation-level high-frequency noise, yielding better-conditioned score distributions.

Point-F1 and Continuous Ranking (PR-AUC) Parity: While latent prediction suppresses nominal false alarms, it does not provide a statistically distinguishable improvement in overall detection fidelity. On the 19 non-GHL streams, TS-JEPA achieves a stream-macro Point-F1 of 0.254 and PR-AUC of 0.333 (dataset-macro Point-F1 of 0.153), compared to 0.207 Point-F1 and 0.336 PR-AUC for TimesNet (dataset-macro Point-F1 of 0.104), and 0.131 Point-F1 and 0.291 PR-AUC for TranAD (dataset-macro Point-F1 of 0.070). Across all 33

TABLE III

CONTROLLED SAME-BACKBONE EXPERIMENT ISOLATING PREDICTION SPACE UNDER IDENTICAL HYBRIDTCNENCODER ($C = 256, S = 64, d = 32$). REPORTED VALUES ARE DATASET-MACRO MEANS ACROSS FIFTEEN DIVERSE BENCHMARK STREAMS ACROSS TEN DOMAINS (NASA SMAP: P-3, A-1; NASA MSL: M-1, C-1; SMD: MACHINE-1-2, MACHINE-1-3; PSM: DEFAULT; SWAN: SF; GECCO: WATER_QUALITY; DAPHNET: S01R01E1, S02R01E0; CALIT2: TRAFFIC; ROOM OCCUPANCY: DEFAULT; GENESIS: DEFAULT; OPPORTUNITY: S1-ADL2) UNDER CLEAN TELEMETRY AND ADDITIVE JITTER NOISE INJECTION ($\text{SNR} \in \{\infty, 20 \text{ dB}, 10 \text{ dB}, 0 \text{ dB}\}$).

Model Backbone	Noise SNR	Val γ_2	Test γ_2	Empirical FPR	Mean False Pos.
TCN-Obs-Recon	Clean (∞)	6.09	57.26	27.29%	10,554.7
	20 dB	6.09	57.23	27.47%	10,587.1
	10 dB	6.09	57.14	32.08%	11,168.1
	0 dB	6.09	57.09	54.10%	17,318.2
TCN-Obs-Pred	Clean (∞)	6.00	55.66	27.57%	10,455.2
	20 dB	6.00	56.09	27.93%	10,505.6
	10 dB	6.00	56.00	32.46%	11,025.9
	0 dB	6.00	55.66	50.98%	15,558.5
TS-JEPA (Ours)	Clean (∞)	0.88	1.77	11.80%	3,949.5 ($\downarrow 62.6\%$)
	20 dB	0.88	1.67	12.25%	4,061.3
	10 dB	0.88	1.54	12.72%	4,248.9
	0 dB	0.88	1.53	13.94%	4,269.0

TABLE IV

NATIVE-WINDOW BASELINE EVALUATION ACROSS FIVE DIVERSE BENCHMARK STREAMS (NASA SMAP: P-3, SMD: MACHINE-1-2, NASA MSL: M-1, DAPHNET: S01R01E1, GECCO: WATER_QUALITY). SHORTENING THE CONTEXT WINDOW TO NATIVE SIZES ($W = 10$ FOR TRANAD, $W = 96$ FOR TIMESNET) EXACERBATES RESIDUAL TAIL HEAVINESS ($\gamma_2 = 448.25$ AND 98.14) RATHER THAN CURING IT.

Model Configuration	Window W	Test γ_2	Mean False Pos.	Empirical FPR
TimesNet (Native)	96	98.14	26,526.8	44.09%
TimesNet (Long)	320	32.04	27,041.0	51.26%
TranAD (Native)	10	448.25	18,765.4	24.51%
TranAD (Long)	320	29.93	16,671.0	31.76%
TS-JEPA (Ours)	320	3.14	10,678.8	22.57%

streams, paired differences between TS-JEPA and TimesNet are $+0.018$ in Point-F1 (95% CI $[-0.052, +0.089]$, $p = 0.55$) and $+0.002$ in PR-AUC ($[-0.098, +0.097]$, $p = 0.30$). Against TranAD, the paired difference is $+0.062$ in Point-F1 ($[-0.012, +0.150]$, $p = 0.94$) and $+0.027$ in PR-AUC ($[-0.075, +0.135]$, $p = 0.30$). Thus, continuous ranking fidelity and calibrated Point-F1 remain statistically comparable between unconstrained latent prediction and observation-space reconstructive baselines.

Event Recall and Detection Conservatism: The lower false-positive rate of latent predictive models is accompanied by a reduction in event recall. On the 19 non-GHL streams, TimesNet captures 58.6% of ground-truth anomaly events, and TranAD captures 48.1%, whereas TS-JEPA captures 49.8%, Reynolds-Stress captures 39.8%, and Potential-Flow captures 33.1%. Latent-space residuals act as a more conservative detector: by filtering high-frequency variations, subtle anomaly onsets that do not produce large latent coordinate shifts are missed, suppressing false alarms at the expense of lower event sensitivity.

Domain-Specific Telemetry Dynamics: Tables VI and VII report the per-dataset breakdown of unadjusted Point-F1 and PR-AUC across the seven evaluated benchmarks. Relative performance depends on the underlying physical sensing regime:

- 1) **Periodic Kinematics (Daphnet):** On wearable Parkinson-

nian accelerometry (6 streams), TimesNet leads in Point-F1 (0.245) and PR-AUC (0.343) over TS-JEPA (0.155 F1, 0.137 PR-AUC). The 2D Fourier representation in TimesNet captures multi-frequency gait periodicity that standard 1D latent projections compress.

- 2) **Spacecraft Telemetry (NASA MSL and SMAP):** On spacecraft telemetry, latent prediction leads. On SMAP (6 streams), TS-JEPA achieves 0.441 Point-F1 and 0.539 PR-AUC, while Operator-Entropy reaches 0.427 F1 and 0.591 PR-AUC, outperforming TimesNet (0.304 F1, 0.463 PR-AUC) and TranAD (0.166 F1, 0.350 PR-AUC). On MSL (4 streams), Operator-Entropy (0.281 F1, 0.445 PR-AUC) and TS-JEPA (0.257 F1, 0.427 PR-AUC) outperform TimesNet (0.159 F1, 0.310 PR-AUC).
- 3) **Solar Flare Magnetograms (SWAN):** On SWAN solar telemetry, TS-JEPA achieves 0.191 Point-F1 and 0.519 PR-AUC, whereas TimesNet triggers zero test detections (0.000 F1, 0.292 PR-AUC) due to severe threshold miscalibration.
- 4) **Quiescent and Heavy-Imbalance Streams (GHL and Metro):** On Gasoil Heating Loop (14 streams) and Metro traffic, all models struggle with near-zero Point-F1 (0.000–0.022). In GHL, anomalies consist of subtle actuator drifts amidst stationary background telemetry; in Metro, extreme traffic volume surges produce out-of-distribution residuals that evade static EVT bounds.

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

A central hypothesis evaluated in this investigation is whether incorporating physics-inspired inductive regularizers into latent representations (Koopman spectral entropy, Reynolds stress closure, and potential-flow curl-free dynamics) improves anomaly detection performance over an unconstrained latent predictive architecture (TS-JEPA).

The empirical data across all 33 streams refute this hypothesis:

- 1) **Operator-Entropy vs. TS-JEPA:** Across all 33 streams (seeds averaged), Operator-Entropy yields a mean paired difference of -0.007 in Point-F1 (95% bootstrap CI $[-0.028, +0.012]$, Wilcoxon $p = 0.71$) and $+0.008$ in PR-AUC ($[-0.013, +0.034]$, $p = 0.28$). Held-out nominal FPR difference is $+0.006$ ($[-0.024, +0.042]$, $p = 0.39$). Across 33 streams, Operator-Entropy outperforms TS-JEPA in Point-F1 on only 9 streams.
- 2) **Reynolds-Stress vs. TS-JEPA:** Reynolds-Stress yields a paired difference of -0.026 in Point-F1 ($[-0.070, +0.004]$, $p = 0.33$) and a statistically significant reduction in PR-AUC (-0.009 , $[-0.024, +0.007]$, $p = 0.013$). Held-out FPR difference is $+0.003$ ($[-0.026, +0.040]$, $p = 0.94$). Reynolds-Stress leads TS-JEPA in F1 on 11 of 33 streams.
- 3) **Potential-Flow vs. TS-JEPA:** Potential-Flow performs significantly worse than unconstrained TS-JEPA across both calibrated Point-F1 (paired difference -0.091 , 95% CI $[-0.178, -0.019]$, $p = 0.020$) and continuous ranking PR-AUC (paired difference -0.076 , 95% CI

TABLE V

HELD-OUT CALIBRATION BENCHMARK ACROSS 33 CONTINUOUS TELEMETRY STREAMS (7 PHYSICAL MODALITIES). THRESHOLDS ARE CALIBRATED STRICTLY ON HELD-OUT NOMINAL VALIDATION SPLITS USING STREAMING PEAKS-OVER-THRESHOLD (SPOT, $q = 10^{-3}$) WITHOUT ACCESS TO TEST ANOMALIES. WE REPORT POINT-F1, THRESHOLD-FREE CONTINUOUS RANKING (PR-AUC), OUT-OF-SAMPLE NOMINAL FALSE-POSITIVE RATE (HELD-OUT FPR, MEASURED ON NOMINAL DATA THE THRESHOLD NEVER SAW), AND EVENT RECALL. “NON-GHL” DENOTES THE STREAM-MACRO MEAN OVER 19 STREAMS (DAPHNET, MSL, SMAP, GENESIS, METRO, SWAN) AVERAGED ACROSS THREE SEEDS. “DATASET-MACRO” IS THE UNWEIGHTED AVERAGE ACROSS THE 7 DATASET MEANS. THE BEST RESULT IN EACH COLUMN IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED.

Model Architecture	Non-GHL Streams (19 Streams, 3 Seeds)				Summary	Gasoil Heating Loop (14 Streams)	
	Point-F1 $\uparrow$	PR-AUC $\uparrow$	Held-Out FPR $\downarrow$	Event Recall $\uparrow$	DS-Macro F1 $\uparrow$	Point-F1 $\uparrow$	Held-Out FPR $\downarrow$
<i>Latent-Space Predictive Architectures (JEPAs)</i>							
TS-JEPA (Unconstrained Control)	0.254	<u>0.333</u>	0.066	<u>0.498</u>	0.153	0.000	0.010
Operator-Entropy JEPA	<u>0.241</u>	0.348	0.079	<u>0.498</u>	<u>0.134</u>	0.001	<u>0.005</u>
Reynolds-Stress JEPA	0.208	0.321	<u>0.076</u>	<u>0.398</u>	<u>0.119</u>	0.001	0.004
Potential-Flow JEPA	0.095	0.205	0.096	0.331	0.065	0.000	<u>0.005</u>
<i>Observation-Space Reconstructive Baselines</i>							
TimesNet (ICLR 2023)	0.207	0.336	0.297	0.586	0.104	0.022	0.058
TranAD (VLDB 2022)	0.131	0.291	0.309	0.481	0.070	<u>0.021</u>	0.049

TABLE VI

PER-DATASET UNADJUSTED POINT-F1 ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED STREAMS) UNDER HELD-OUT CALIBRATION (SPOT, $q = 10^{-3}$). BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND SECOND BEST IS UNDERLINED. THE BOTTOM ROW REPORTS UNWEIGHTED DATASET-MACRO AVERAGES MATCHING TABLE V.

Dataset	Physical Sensing Domain	Streams	TS-JEPA (Control)	Operator-Entropy	Reynolds-Stress	Potential-Flow	TimesNet	TranAD
Daphnet	Wearable Parkinsonian Kinematics	6	<u>0.155</u>	0.134	0.135	0.066	0.245	0.149
GHL	Gasoil Heating Loop Dynamics	14	0.000	0.001	0.001	0.000	0.022	<u>0.021</u>
Genesis	Pick-and-Place Automation	1	0.028	<u>0.012</u>	0.000	0.000	0.000	0.000
MSL	NASA Mars Science Laboratory	4	0.257	0.281	<u>0.264</u>	0.115	0.159	0.152
SMAP	NASA Soil Moisture Active Passive	6	0.441	<u>0.427</u>	0.331	0.137	0.304	0.166
Metro	Interstate Highway Traffic (I-94)	1	0.000	0.000	0.001	0.005	<u>0.001</u>	0.000
SWAN	Solar Flare Magnetic Telemetry	1	0.191	0.081	0.102	<u>0.132</u>	0.000	0.000
Dataset-Macro Mean	Across All 7 Benchmarks	33	0.153	<u>0.134</u>	0.119	0.065	0.104	0.070

TABLE VII

PER-DATASET PRECISION-RECALL AREA UNDER THE CURVE (PR-AUC / AVERAGE PRECISION) ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED CHANNELS). THE BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED. CONTINUOUS RANKING EVALUATES THE UNTHRESHOLDED PREDICTIVE SCORES.

Dataset	Physical Telemetry Domain	Streams	TS-JEPA (Control)	Operator-Entropy	Reynolds-Stress	Potential-Flow	TimesNet	TranAD
Daphnet	Wearable Parkinsonian Kinematics	6	0.137	0.150	0.151	0.144	0.343	<u>0.280</u>
GHL	Gasoil Heating Loop Dynamics	14	0.016	<u>0.015</u>	0.011	0.013	0.009	0.010
Genesis	Pick-and-Place Automation	1	0.053	0.009	0.005	0.005	<u>0.020</u>	0.006
MSL	NASA Mars Science Laboratory	4	0.427	0.445	<u>0.413</u>	0.281	0.310	0.294
SMAP	NASA Soil Moisture Active Passive	6	<u>0.539</u>	0.591	0.522	0.261	0.463	0.350
Metro	Interstate Highway Traffic (I-94)	1	0.001	0.001	0.001	0.001	<u>0.002</u>	0.002
SWAN	Solar Flare Magnetic Telemetry	1	<u>0.519</u>	0.370	0.397	0.325	0.292	0.562
Dataset-Macro Mean	Across All 7 Benchmarks	33	0.242	<u>0.227</u>	0.214	0.147	0.206	0.216

$[-0.142, -0.019]$, $p = 0.039$). Potential-Flow outperforms TS-JEPA on only 5 of 33 streams.

These findings are corroborated by the matched-capacity regularizer controls reported in Table II(a). When the underlying neural architecture is held fixed on the seven-stream multi-seed subset:

- Disabling the spectral entropy loss ($\lambda_{\text{ent}} = 0$) changes Point-F1 by $+0.001$ ($[-0.006, +0.009]$, $p = 1.00$).
- Disabling the Reynolds stress loss ($\alpha_{\text{stress}} = 0$), or scoring via predictive mean error alone, produces differences that do not exceed seed noise ($p \geq 0.375$).
- Disabling Grassmannian regime routing in Potential-Flow changes Point-F1 by $+0.006$ ($[-0.022, +0.032]$, $p = 0.58$).

We conclude that the physical regularizers do not provide a measurable anomaly detection advantage over unconstrained latent prediction. Whatever performance differences exist be-

tween models arise from shared representation conditioning rather than the physical loss terms.

C. Mathematical Diagnostics of Physical Regularizers

To understand why the proposed regularizers enforce their intended mathematical properties yet fail to improve downstream anomaly detection, we conduct targeted dynamical diagnostic experiments.

1) *Operator-Entropy: Anti-Collapse Spectral Rank Preservation:* In Operator-Entropy JEPA, the transition operator $K \in \mathbb{R}^{D \times D}$ ($D = 32$) is regularized via Von Neumann spectral entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$ on normalized Gram matrix $\rho = KK^T / \text{Tr}(KK^T)$. Table VIII evaluates effective operator rank via the RankMe metric ($\exp(\mathcal{H})$ where $\mathcal{H}$ is the spectral entropy of singular values), representation covariance conditioning, and downstream calibrated Point-F1 across five telemetry streams.

TABLE VIII

EMPIRICAL SPECTRAL DIAGNOSTICS OF LEARNED TRANSITION OPERATORS AND REPRESENTATIONS ACROSS FIVE BENCHMARK TELEMETRY STREAMS ($D = 32$). OPERATOR-ENTROPY PRESERVES FULL THEORETICAL RANK ($\exp(\mathcal{H}) = 31.996$ OUT OF 32.0), PREVENTING DIMENSIONAL COLLAPSE.

Model Architecture	Operator Rank (RankMe)	Rep. Rank	Point-F1	Empirical FPR
Operator-Entropy JEPA	31.996	3.564	0.0710	0.1497
Representation-Entropy JEPA	24.647	7.579	0.0479	0.0442
Vanilla TS-JEPA (Unconstrained)	24.807	3.993	0.0301	0.0371
VICReg TS-JEPA	24.339	5.881	0.0043	0.0273

TABLE IX

CONTROLLED COVARIANCE VS. MARGINAL ANOMALY BENCHMARK. REYNOLDS-STRESS JEPA IMPROVES MARGINAL VARIANCE DETECTION OVER UNCONSTRAINED TS-JEPA (0.5838 vs. 0.3313 PR-AUC), BUT BOTH LATENT MODELS STRUGGLE ON PURE CORRELATION FLIPS.

Model Architecture	Pure Covariance Shift		Pure Marginal Shift	
	PR-AUC	Point-F1	PR-AUC	Point-F1
Reynolds-Stress JEPA	0.0498	0.0000	0.5838	0.0593
TS-JEPA (Unconstrained)	0.0409	0.0000	0.3313	0.0000
TimesNet	0.9313	0.6558	0.9998	0.6267
TranAD	0.6761	0.7246	0.9343	0.6560

As shown in Table VIII, unconstrained Vanilla TS-JEPA and VICReg collapse across 7–8 singular modes, yielding an effective operator rank of 24.3–24.8. Operator-Entropy JEPA achieves an effective operator rank of 31.996 out of 32.0, confirming that the regularizer prevents modal collapse. However, full operator rank does not improve downstream anomaly detection: telemetry anomalies are frequently localized subsystem deviations occurring in low-dimensional subspaces. Preserving high-rank transition dynamics forces the model to track ambient high-frequency variations, which does not enhance fault discrimination.

2) *Reynolds-Stress: Covariance Alignment vs. Marginal Perturbations*: Reynolds-Stress JEPA aligns second-order latent fluctuation statistics ($\Sigma_{z'} = \frac{1}{B} \sum z'(z')^T$) between predicted and target horizons. Table IX compares models on synthetic telemetry isolating pure cross-channel correlation flips ($\rho : +0.85 \rightarrow -0.85$) from pure marginal variance scaling ($3.0\times$).

Table IX indicates that on marginal variance shifts, Reynolds-Stress improves PR-AUC by +76% over TS-JEPA (0.5838 vs. 0.3313). However, on pure covariance shifts without marginal changes, both latent models show low sensitivity (0.041–0.050 PR-AUC) relative to observation-space baselines (0.676–0.931). Standard temporal pooling across channels averages out instantaneous phase alignments, preventing the second-moment loss from isolating cross-channel relationship shifts unless explicit pairwise relational graphs are modeled.

3) *Potential-Flow: Conservative vs. Circulating Dynamics*: Potential-Flow JEPA constrains transition velocities to curl-free potential gradients ($\hat{v} = -\nabla\Phi$). Table X evaluates Potential-Flow against Helmholtz-JEPA (decomposing velocity into gradient plus solenoidal circulation: $\hat{v} = -\nabla\Phi + \text{curl}(A)$) and unconstrained TS-JEPA across two dynamical benchmarks: a conservative Duffing oscillator and a non-conservative Van der Pol limit-cycle attractor.

Table X confirms that pure curl-free potential flow is math-

TABLE X

DYNAMICAL SYSTEM VALIDATION COMPARING CURL-FREE POTENTIAL-FLOW ($\hat{v} = -\nabla\Phi$), HELMHOLTZ DECOMPOSITION ($\hat{v} = -\nabla\Phi + \text{curl}(A)$), AND UNCONSTRAINED TS-JEPA ON CONSERVATIVE HAMILTONIAN VS. NON-CONSERVATIVE LIMIT-CYCLE PHASE SPACES.

Model Architecture	Conservative Duffing		Non-Conservative Van der Pol	
	PR-AUC	Point-F1	PR-AUC	Point-F1
Helmholtz JEPA ($\nabla\Phi + \text{curl}(A)$)	0.5985	0.7815	0.5250	0.7574
TS-JEPA (Unconstrained)	0.5855	0.7583	0.4577	0.6660
Potential-Flow JEPA ($-\nabla\Phi$)	0.1701	0.2036	0.3106	0.3548

ematically incapable of representing closed circulating orbits, yielding poor detection on Van der Pol (0.3548 Point-F1 vs. 0.7574 for Helmholtz JEPA). Because physical telemetry often exhibits non-conservative, cyclic behavior (e.g., diurnal cycles, rotational turbines), enforcing irrotational dynamics severely penalizes normal circulating trajectories, explaining Potential-Flow’s poor benchmark performance (0.065 dataset-macro F1 in Table V).

D. Operational Incident Evaluation: Event Metrics and Adaptation

Evaluating telemetry solely by point-level metrics can penalize slight temporal boundary offsets in continuous anomalies. In operational systems, faults represent extended continuous incidents; a detector that identifies the event with a minor phase lag is penalized on every out-of-boundary point.

To analyze operational incident detection, Table XI evaluates three complementary metrics across three telemetry streams: Room Occupancy (stream: default, 11 events), NASA SMAP (stream: P-3, 1 event), and CalIt2 (stream: traffic, 21 events). We compare strict Point-F1, Event-F1 ($P_{\text{event}}, R_{\text{event}}$), and Tatbul et al. Range-Aware F1 ($F1_{\text{range}}$) across three operational inference configurations: Standard (middle-window, static EVT), Sharp (dense causal, static EVT), and Adaptive (dense, dynamic rolling SPOT).

Table XI highlights three operational characteristics: First, static EVT thresholding achieves 100% event recall on Room Occupancy and SMAP (P-3), but generates over 4,000 false alarm points on Room Occupancy due to unmodeled diurnal baseline shifts, causing point precision to collapse to 20.0%. Second, dynamic rolling SPOT suppresses false alarms by $21\times$ (from 4,705 to 223 points) and raises precision to 29.9%, but event recall drops from 100% to 63.6%. Tracking non-stationary drift via rolling quantiles creates an operational trade-off between alarm fatigue and missed events. Third, on transient burst streams (CalIt2, 21 short spikes), models detect only 4.8%–23.8% of events under rolling SPOT, demonstrating that short pulses remain difficult to distinguish from ambient noise.

E. Operational Latency: Strictly Causal Online Detection vs. Buffered Horizons

Sliding-window anomaly detection algorithms typically evaluate a target suspect region S , incurring an effective lookahead delay of S steps ($S = 64$ in our core protocol).

TABLE XI

OPERATIONAL INCIDENT DETECTION COMPARISON ACROSS REPRESENTATIVE TELEMETRY STREAMS: ROOM OCCUPANCY (STREAM: `DEFAULT`, 11 GROUND-TRUTH EVENTS), NASA SMAP (STREAM: `P-3`, 1 THRUSTER FAULT), AND CALIT2 (STREAM: `TRAFFIC`, 21 PEDESTRIAN BURST EVENTS). WE CONTRAST STANDARD (MIDDLE, STATIC EVT), SHARP (DENSE, STATIC EVT), AND ADAPTIVE (DENSE, DYNAMIC SPOT) INFERENCE MODES.

Evaluated Stream	Operational Configuration	Pt-F1	Pt-Prec	Evt-F1	Evt-Prec	Evt-Rec	Rng-F1	False Pos.
<i>Room Occupancy: Stream default (Environmental Sensing, 11 Ground-Truth Incidents)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.3728	22.9%	0.5238	35.5%	100%	0.3708	3,842
operator_entropy	Sharp (Dense, Static EVT)	0.3329	20.0%	0.9167	84.6%	100%	0.3329	4,705
operator_entropy	Adaptive (Dense, Dynamic SPOT)	0.1273	29.9%	0.6364	63.6%	63.6%	0.3699	223 ($\downarrow 21\times$)
potential_flow	Sharp (Dense, Static EVT)	0.3431	20.7%	0.9167	84.6%	100%	0.3431	4,496
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0441	10.2%	0.5714	60.0%	54.5%	0.1636	291 ($\downarrow 15\times$)
<i>NASA SMAP: Stream P-3 (Spacecraft Telemetry, 1 Continuous Fault Segment)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.0087	14.6%	0.1818	10.0%	100%	0.2266	35
operator_entropy	Sharp (Dense, Static EVT)	0.1274	27.3%	0.1538	8.3%	100%	0.3634	295
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0110	6.9%	0.3333	20.0%	100%	0.1213	108
<i>Callit2: Stream traffic (Pedestrian Entrance Flow, 21 Transient Event Clusters)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.0131	1.9%	0.0588	7.7%	4.8%	0.0216	53
operator_entropy	Adaptive (Dense, Dynamic SPOT)	0.1368	8.7%	0.3226	50.0%	23.8%	0.1274	357
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0903	14.3%	0.0909	100.0%	4.8%	0.0714	42

In mission-critical streaming applications, algorithms must operate under strictly causal conditions ($\tau = 0$ lookahead).

Table XII reports performance across five matched telemetry streams (NASA SMAP `P-3`, SMD machine-1-2, NASA MSL `M-1`, Daphnet `S01R01E1`, and GECCO `water_quality`) under strictly causal (0-lookahead) versus buffered (64-lookahead) scoring.

Under strictly causal streaming, TS-JEPA maintains an empirical false-positive rate of 4.18% (41.76 false alarms per 1,000 points), generating an average of 1,685.6 false alarms per stream. In contrast, TimesNet and TranAD generate 16,952.0 and 16,637.6 false alarms (empirical FPRs of 30.06% and 20.81%). On GECCO water telemetry (113,000 test points), TimesNet and TranAD trigger over 81,500 false alarms (99.6% FPR), whereas TS-JEPA triggers zero false alarms.

Two methodological trade-offs govern these results: First, the false-alarm reduction under strict causality comes at the expense of early event capture. TS-JEPA detects 10.7% of anomaly events within $\tau \leq 64$ steps under causal streaming, compared to 22.9% for TimesNet and 32.9% for TranAD. Second, when strictly causal inference is evaluated under held-out calibration across the full 33 streams, measured nominal FPR drops to 0.47%–1.13% for the JEPA family (vs. 20.14%–22.02% for baselines), but Point-F1 reverses: TS-JEPA reaches 0.1132 ± 0.0447 , Operator-Entropy reaches 0.1018 ± 0.0041 , Reynolds-Stress reaches 0.0697 ± 0.0406 , and Potential-Flow reaches 0.0275 ± 0.0161 , compared to 0.1983 ± 0.0194 for TimesNet and 0.1321 ± 0.0164 for TranAD. Strictly causal latent scoring therefore represents an operational choice that suppresses false alarms at the cost of lower point-level F1 and delayed event recall.

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

A critical requirement for autonomous telemetry monitoring is whether automated extreme-value thresholding adheres to theoretical design risk parameters. When practitioners specify

a design risk of $q = 10^{-3}$ (0.1%), the operational expectation is that nominal false alarms will occur at or below this frequency.

Table XIII reports a controlled comparison of threshold calibrators evaluated on identical score streams across the 33 continuous telemetry streams under buffered inference. The empirical results demonstrate several key realities:

- 1) **Systematic Design-Risk Overshoot:** Extreme-value thresholding via SPOT ($u = 0.98, q = 10^{-3}$) systematically fails to achieve its design risk on out-of-sample nominal telemetry. Across all 33 streams, empirical nominal FPR averages 5.17% for TS-JEPA, 6.15% for Operator-Entropy, 23.88% for TimesNet, and 24.59% for TranAD, overshooting the theoretical 0.1% target by 50-fold to 245-fold.
- 2) **High Failure Rate Across Streams:** The 95% bootstrap confidence intervals for held-out nominal FPR strictly exclude the theoretical design risk $q = 10^{-3}$ in 57% of TS-JEPA runs, 65% of Operator-Entropy runs, 87% of TimesNet runs, and 84% of TranAD runs. Real-world telemetry exhibits non-stationary baseline drift and seasonal fluctuations that violate the stationary tail assumptions of Extreme Value Theory.
- 3) **Parity with Simple Empirical Quantiles:** Simple non-parametric quantile rules match or exceed the performance of SPOT. On TS-JEPA, taking the validation maximum yields an empirical held-out FPR of 0.0524 and Point-F1 of 0.1922, while an empirical 99.5th percentile rule yields an FPR of 0.0657 and Point-F1 of 0.1942. Parametric GPD tail modeling provides no demonstrable benefit over simple empirical percentiles on these benchmarks.
- 4) **Failure of Kurtosis-Adaptive Calibration:** The kurtosis-adaptive multiplier rule ($\tau_{\text{eff}} = \tau(1 + \frac{1}{6} \tanh \gamma_2)$) substantially degrades nominal calibration. On TS-JEPA, it inflates held-out nominal FPR from 0.0517 to 0.1880 (with CI excluding q rising from 57% to 84%), while yielding an indistinguishable Point-F1 (0.1947 vs. 0.1929). On

TABLE XII

OPERATIONAL LATENCY AND DETECTION FIDELITY UNDER STRICTLY CAUSAL (0-LOOKAHEAD DELAY) VERSUS BUFFERED WINDOW (64-LOOKAHEAD DELAY) INFERENCE ACROSS 5 REPRESENTATIVE TELEMETRY STREAMS. UNDER CAUSAL STREAMING, TS-JEPA ACHIEVES AN EMPIRICAL FALSE-POSITIVE RATE OF 4.18% ($\text{FAR}_{1k} = 41.76$ FALSE ALARMS PER 1K POINTS), COMPARED TO 30.06% FOR TIMESNET AND 20.81% FOR TRANAD.

Inference Setting	Model	Empirical FPR	FAR_{1k} (/1k pts)	False Positives	Point-F1	Precision	Mean Delay (Δt)	Recall ($\tau \leq 64$)
Strictly Causal (0-Lookahead)	TS-JEPA	0.0418	41.76	1,685.6	0.0688	0.1581	130.3	0.1071
	TimesNet	0.3006	300.63	16,952.0	0.0774	0.1525	251.2	0.2286
	TranAD	0.2081	208.13	16,637.6	0.0881	0.1255	43.2	0.3286
Buffered Window (64-Lookahead)	TS-JEPA	0.1514	151.39	2,696.0	0.0431	0.0355	89.6	0.1357
	TimesNet	0.2800	280.03	16,916.8	0.0807	0.1404	235.3	0.2571
	TranAD	0.1998	199.82	16,339.6	0.0197	0.2284	32.8	0.2786

TABLE XIII

CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS ACROSS 33 CONTINUOUS TELEMETRY STREAMS UNDER BUFFERED INFERENCE. THEORETICAL DESIGN RISK FOR SPOT IS $q = 10^{-3} = 0.1\%$. THE SHARE OF RUNS WHOSE 95% BOOTSTRAP CONFIDENCE INTERVAL EXCLUDES $q = 10^{-3}$ DEMONSTRATES THAT AUTOMATED CALIBRATION SYSTEMATICALLY OVERSHOTS ON OUT-OF-SAMPLE TELEMETRY.

Model Architecture	Calibrator Rule	Held-Out Nominal FPR	CI Excludes $q = 10^{-3}$	Point-F1
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	0.0517	57%	0.1929
	SPOT + Kurtosis ($\beta = 0.167$)	0.1880	84%	0.1947
	Empirical 99.5th Percentile	0.0657	47%	0.1942
	Validation Maximum	0.0524	60%	0.1922
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	0.0615	65%	0.1835
	SPOT + Kurtosis ($\beta = 0.167$)	0.1098	76%	0.1666
	Empirical 99.5th Percentile	0.0752	60%	0.1903
	Validation Maximum	0.0691	63%	0.1869
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.2388	87%	0.1614
	SPOT + Kurtosis ($\beta = 0.167$)	0.2439	85%	0.1550
	Empirical 99.5th Percentile	0.2488	91%	0.1626
	Validation Maximum	0.2430	87%	0.1623
TranAD	Standard SPOT ($q = 10^{-3}$)	0.2459	84%	0.1039
	SPOT + Kurtosis ($\beta = 0.167$)	0.2659	83%	0.1047
	Empirical 99.5th Percentile	0.2616	88%	0.1062
	Validation Maximum	0.2535	85%	0.1054

TABLE XIV

COMPUTATIONAL PROFILE, MEMORY CONSUMPTION, LATENCY, AND THROUGHPUT BENCHMARKED ON AN NVIDIA RTX 5090 GPU ($K = 25$, WINDOW SIZE 320).

Model Architecture	Parameters	VRAM (MB)	Latency (ms/win)	Infer (win/s)	Train (win/s)
TS-JEPA	289,061	183.8	0.6072	1,646.8	612.8
Operator-Entropy JEPA	285,733	183.9	0.6939	1,441.0	553.2
Reynolds-Stress JEPA	295,953	184.0	0.7456	1,341.2	535.8
Potential-Flow JEPA	338,117	184.4	0.8445	1,184.2	400.2
Helmholtz JEPA	416,391	185.3	0.8378	1,193.6	526.6
TimesNet	648,553	279.6	0.2908	3,439.4	1,161.4
TranAD	205,746	1,280.0	0.1897	5,272.2	1,554.4

Operator-Entropy, it raises nominal FPR from 0.0615 to 0.1098. Excess kurtosis measured in training splits does not accurately forecast non-stationary tail shifts on out-of-sample telemetry.

G. Computational Profile: Complexity, Latency, and Memory Footprint

Table XIV benchmarks parameter complexity, peak GPU memory footprint (VRAM), forward inference latency, and streaming throughput across all evaluated architectures on an NVIDIA RTX 5090 GPU (input dimension $K = 25$, batch size 32, context $C = 256$, suspect $S = 64$).

As shown in Table XIV, Operator-Entropy JEPA possesses fewer trainable parameters (285,733) than unconstrained TS-JEPA (289,061), because its explicit Koopman transition matrix $K \in \mathbb{R}^{32 \times 32}$ (1,024 weights) replaces the multi-layer MLP predictor. All JEPA variants maintain a compact peak GPU memory footprint of 183.8–185.3 MB, consuming

nearly $7\times$ less memory than TranAD (1,280.0 MB), whose self-attention maps scale quadratically with window length. Forward inference latency for the JEPA family ranges between 0.60–0.84 ms per window (1,184–1,647 windows per second). While TimesNet (0.29 ms) and TranAD (0.19 ms) achieve faster forward inference, the JEPA family operates well within sub-millisecond operational streaming deadlines.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we conducted an empirical investigation into whether Joint-Embedding Predictive Architectures (JEPAs) and physics-inspired inductive regularizers resolve the false-alarm vulnerability of multivariate time-series anomaly detection. By evaluating 33 continuous telemetry streams across seven physical sensing domains under a strictly held-out calibration protocol, this study yields three primary conclusions:

First, forward prediction in a contracted latent representation space fundamentally conditions residual distributions. By filtering out localized observation-space sensor jitter, latent prediction reduces test residual excess kurtosis and achieves a statistically significant 4-fold to 5-fold reduction in out-of-sample nominal false-positive rates compared to reconstructive baselines (paired difference -0.154 against TimesNet, $p = 0.008$; -0.157 against TranAD, $p = 0.001$). However, continuous ranking (PR-AUC) and calibrated Point-F1 remain statistically indistinguishable between unconstrained latent prediction and baselines ($p \geq 0.30$). Under strictly causal streaming, latent prediction suppresses false alarms to 0.47%–4.18%, but at the cost of lower point-level F1 and delayed event recall.

Second, empirical hypothesis testing refutes the claim that physics-inspired regularizers provide an anomaly detection advantage over unconstrained latent prediction. Across all 33 streams, neither Operator-Entropy (spectral rank preservation) nor Reynolds-Stress (second-moment fluctuation alignment) outperforms unconstrained TS-JEPA ($p \geq 0.33$), while Potential-Flow (curl-free gradient dynamics) performs significantly worse ($p = 0.020$) due to its inability to represent circulating limit-cycle orbits. In matched-capacity controls holding the neural backbone fixed, disabling each physical loss term changes Point-F1 by $+0.001$ to $+0.006$ ($p \geq 0.58$). While diagnostic experiments confirm that these formulations enforce their intended mathematical properties, real-world telemetry deviations do not obey idealized physical conservation laws, rendering these regularizers ineffective as general anomaly detection priors.

Third, automated threshold calibration via Extreme Value Theory (SPOT) systematically overshoots its theoretical design risk on non-stationary telemetry. At $q = 10^{-3}$, empirical nominal false-positive rates settle at 5.2%–7.4% for JEPAs and 23.9%–24.6% for reconstructive baselines, with 95% bootstrap intervals excluding q in 57%–87% of runs. Parametric EVT tail modeling provides no measurable advantage over simple empirical quantiles, and kurtosis adjustments fail to correct pre-asymptotic tail drift.

Four directions extend this investigation. First, incorporating solenoidal vorticity components into velocity fields via Helmholtz decomposition enables modeling dissipative systems with limit-cycle attractors without sacrificing conservative stability. Second, developing memory-bounded online drift adaptation will allow tracking diurnal thermal cycles and operational setpoint transitions without buffer contamination. Third, implementing accelerated sliding-window inference kernels will enable scaling multi-seed evaluations to million-step telemetry streams. Finally, quantizing JEPA encoders into 8-bit integer formats will support deployment on radiation-hardened satellite microcontrollers and low-power industrial edge hardware.

CODE AND DATA AVAILABILITY

To ensure reproducibility, all benchmark extraction scripts, preprocessing pipelines, model checkpoints, PyTorch training implementations, multi-seed evaluation runners (scripts/run_multiseed_heldout_benchmark.py), protocol ablation runners (scripts/run_protocol_ablation.py), and evaluation logs across all 33 telemetry streams are open-sourced in our public repository: https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21. All public telemetry datasets (NASA SMAP/MSL, Daphnet, GHL, Genesis, Metro Traffic, and SWAN) are accessible via their referenced repositories.

REFERENCES

- [1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.
- [2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.
- [3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.
- [4] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.
- [5] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2023.
- [6] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.
- [7] C. U. Carmona, F.-X. Aubert, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.
- [8] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.
- [9] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.
- [10] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.
- [11] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.
- [12] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.
- [13] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.
- [14] Y. Liu, C. Li, J. Wang, and M. Long, "Koopman: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.
- [15] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.
- [16] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dedector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.
- [17] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.
- [18] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.
- [19] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.
- [20] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.
- [21] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.
- [22] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwok, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.
- [23] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.
- [24] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
- [25] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.
- [26] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.
- [27] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.
- [28] A. Siffer, P.-A. Fouque, A. Termier, and C. Lagouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.
- [29] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

- [30] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," <https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts>, 2007.
- [31] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.
- [32] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.
- [33] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.
- [34] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.
- [35] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.
- [36] R. Wu and E. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2434, 2023.